PROGRESS BRIEFING · OCTOBER 2026

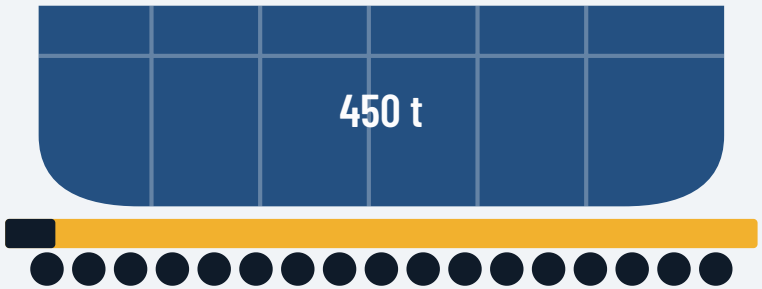
Carry a heavy hull block on one big transporter, or on several small ones coupled together?

Single-carrier capacity or cooperative transport: shipyard transporter fleets sized under one service specification and priced from tender awards

Service-constrained fleet sizing for single-carrier and cooperative block transport in shipyards

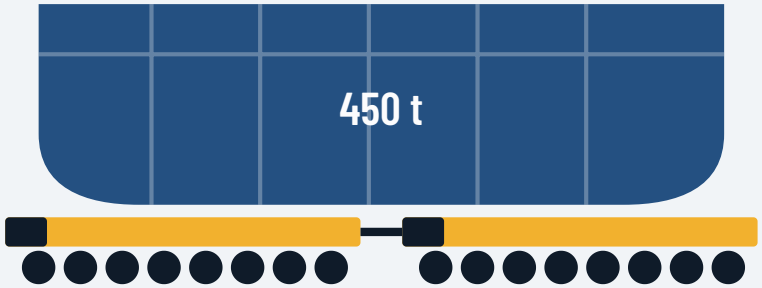
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Target journal: Ocean Engineering (manuscript being finalised)

A · 1 × 500 t



1 vehicle · 4 crew

B · 2 × 270 t



2 vehicles · 8 crew, wait for both, then couple

The whole study in one minute

1

The question

Coupled transporters can move blocks heavier than any single vehicle. Should a yard buy heavy units, light units that couple, or a mix? Existing studies schedule a given fleet.

2

What we did

Nine tiers and light–heavy mixes, each sized as the smallest count found to meet one service standard (≥ 95% on time, none over 120 min late) over 30 days in 36 conditions; priced from 33 tender awards.

3

The answer

Under shift labour the least-cost fleet couples at most one block in ten in **91.0%** of 1,620 decisions, and in all 1,350 where some tier carries every block. In 13 further conditions light units with matched heavy units win 95%.

4

Why

Repeated service and alignment make up a median 90% of a light fleet's extra transporter-time; partner waiting, 2%. Under shift labour a light tier can couple only about 2–18% of its workload.

> **Coupling is a remedy for exceptional blocks, not a purchasing strategy.**

Sources: workload Roh & Cha (2011); block masses Jiang et al. (2021), Roh & Cha (2011), Liu et al. (2022); handling Liu et al. (2022), Kim & Joo (2012); prices 33 public award notices (2022–2026, three procurement platforms); wages National Bureau of Statistics of China (2026), Table 7, and China Statistical Yearbook 2025, table 4-12. Boxes 3 and 4 are results of this study (paper abstract, §5).

The order of this briefing (click to jump)

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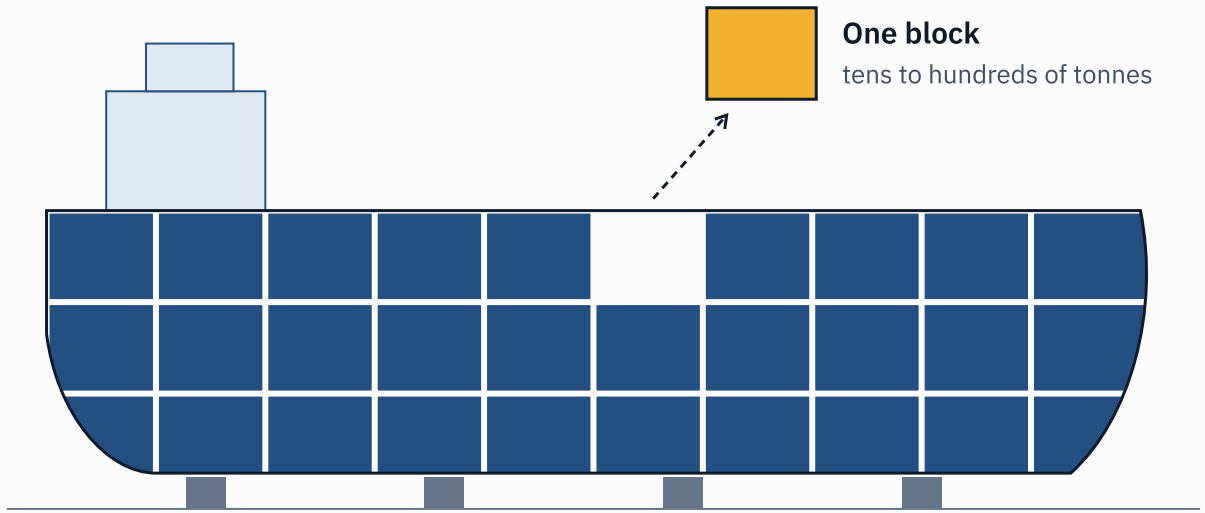
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Ships are built in blocks: blocks are made first and joined in the dock



A hull in the dock, welded together from many blocks (schematic)

- A large ship is divided into many **blocks**, built, outfitted and painted **in parallel** in different shops and joined in the dock
- A typical block is about 15 m × 15 m × 5 m and weighs 100–300 t; some exceed 500 t
- The price: blocks are **moved many times**, and a late move holds up the next process

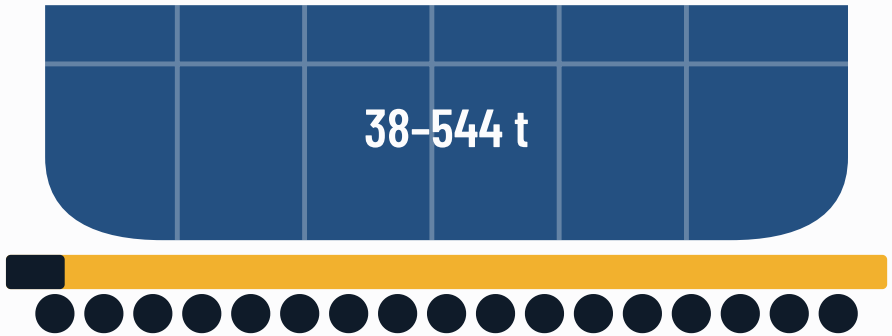
This study is about those moves: which transporters, and how many, move them at least cost.

Sources: block size and mass Park & Seo (2012), p. 390.

Each block is moved several times between shops, yards and the dock

A typical route of one block

Assembly shop → Outfitting shop → Paint shop → Block yard → Assembly yard → Dock

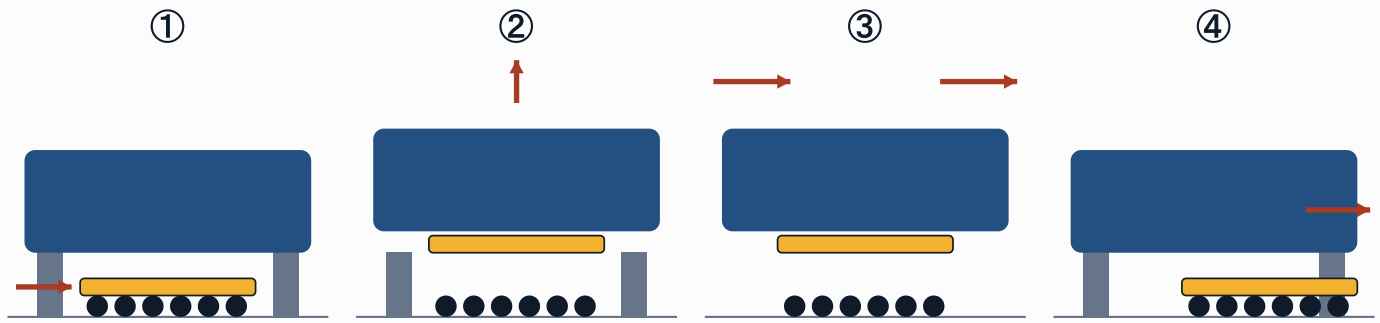


Self-propelled hydraulic transporter, rated 200–550 t

- **Block masses** in published samples: 38–450 t, 204–399 t, 260–544 t
- **Transporters**: self-propelled hydraulic units; fleets in the literature are rated 200–550 t
- **Price**: about CNY 1.79–3.92 million per unit for 200–550 t (ex VAT, fitted to tender awards)
- **Crew**: four or five workers (a driver and guiders) per transporter in practice; this study uses the lower end, a crew of four per transporter (one driver, three guiders), not shared
- **Workload**: on one recorded day four transporters moved 97 blocks in 16 h

Sources: masses Jiang et al. (2021) Tables 5, A1, A2; Liu et al. (2022) S1 data; Roh & Cha (2011) Table 2. Fleet ratings Jiang et al. (2021) Table 5; Roh & Cha (2011) Table 1; Liu et al. (2022) Table 4. Crew Park & Seo (2012), p. 392. Workload Roh & Cha (2011), p. 3246. Prices 33 public tender awards (2022–2026).

How a transporter moves a block: drive under, lift, drive, set down



① Drive in empty
The block rests on supports; the vehicle drives into the gap underneath

② Lift hydraulically
The platform rises and lifts the block off the supports: this is loading

③ Drive loaded, slowly
It carries the block to the destination at about half its empty speed

④ Set down, leave
The platform lowers the block onto new supports (unloading), and the empty vehicle heads for the next block

Rated capacity
Sold by tier; one unit cannot carry a block above its rating

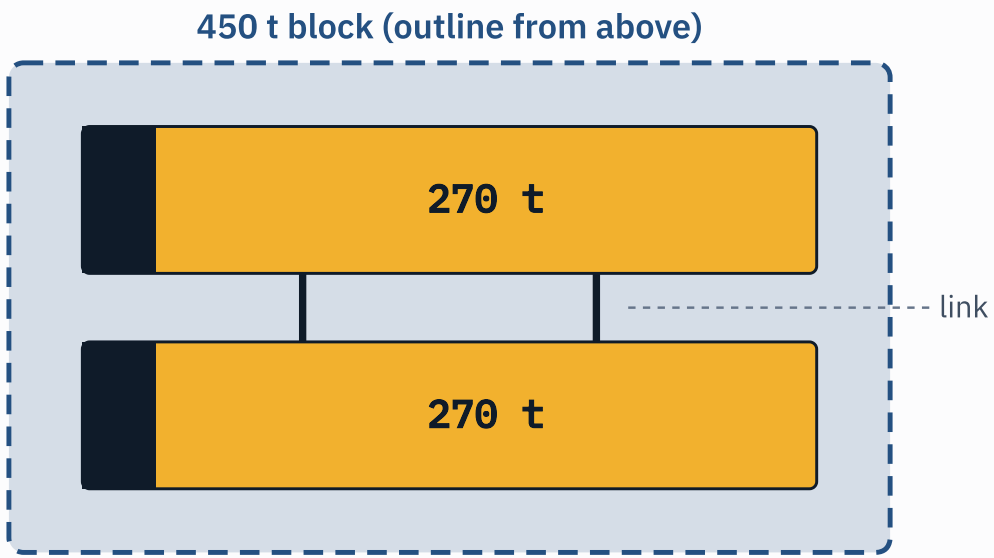
Speed
270 t model: 12 km/h empty, 6 km/h loaded

Loading / unloading
10 min each; a mass-dependent formula also exists; loading in practice often takes 30–60 min

Crew
Four or five per vehicle in practice (a driver and guiders); here each has its own crew of four: one driver, three guiders

Sources: speeds Suzhou Dafang DCY270 data sheet (archived 2026-09-24); 10 min each Liu et al. (2022) §4.1.1; mass-dependent formula Kim & Joo (2012) §4; 30–60 min Liu et al. (2022) §4.2.2; crew Park & Seo (2012), p. 392.

Coupling: two or three units side by side drive as one large vehicle



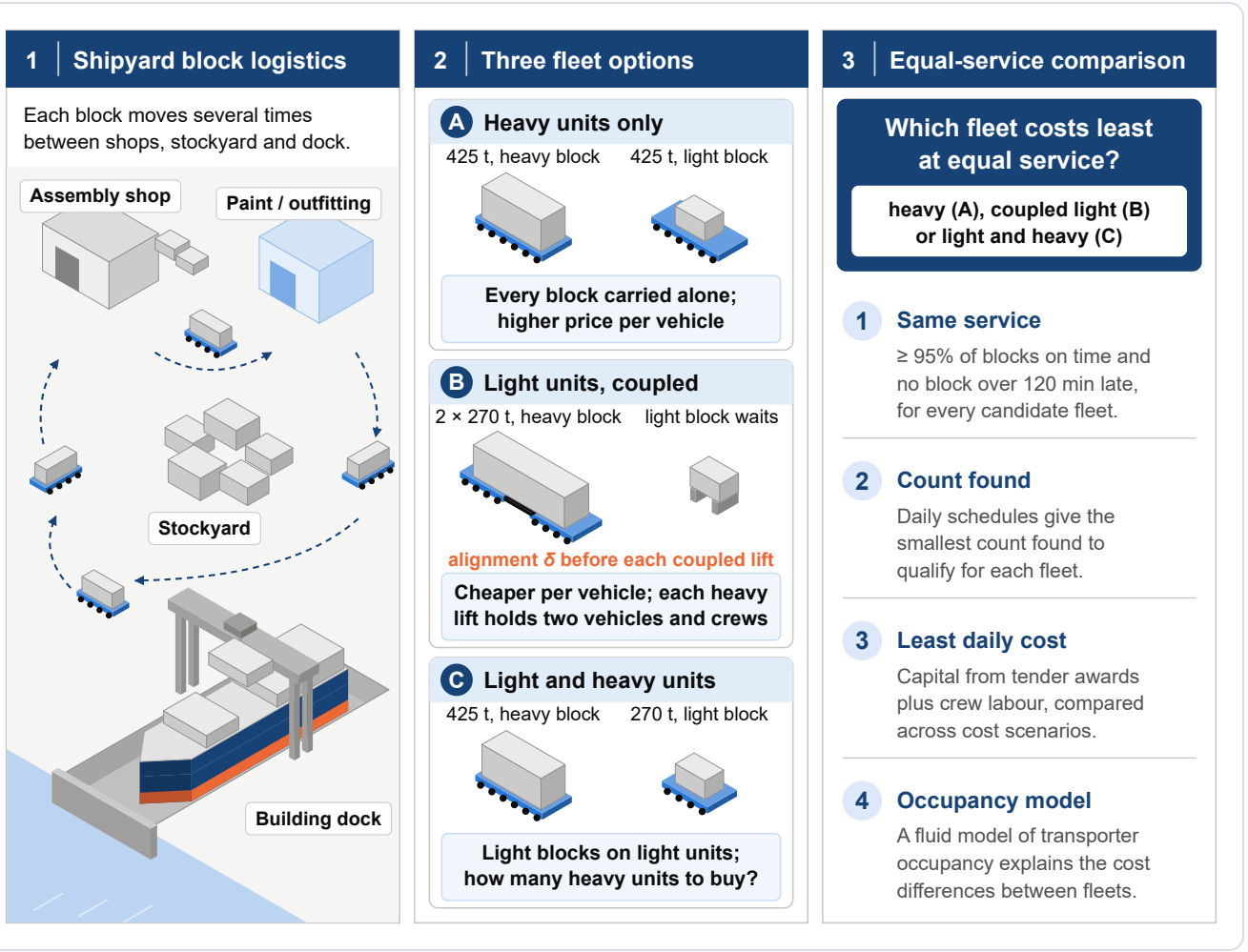
270 + 270 = 540 t ≥ 450 t

- **Capacities add**: two 270 t units lift up to 540 t, three up to 810 t; the set runs under synchronised control
- **All must be there**: the first to arrive waits for the last
- **Alignment takes time**: no measured value is published; we assume 10 min and test 0–40 min
- **Held throughout**: from loading to unloading, each unit and its crew can do nothing else
- At most three units per team (assumed; two also tested)

Coupling lets light units carry heavy blocks, but one block then takes two or three times the vehicles and crews: cheaper capital, more vehicle-time.

Sources: synchronised control TII Group self-propelled transporter brochure (0317-EN), p. 21; big tasks by several flatcars Jiang et al. (2021), constraint (24); coupled-use orders on the Suzhou Dafang website. Alignment time and team size are our assumptions.

The problem in one figure: block logistics, three fleet options, one comparison



- ① **Block logistics**: each block moves several times between shops, stockyard and building dock
- ② **Three fleet options**: A heavy units only; B light units coupled for heavy blocks, with alignment δ before each coupled lift; C light units with a few heavy units, and the question of how many
- ③ **The comparison**: the same service → the smallest count found by scheduling → the least daily cost; a fluid occupancy model explains the cost differences

> **Every fleet is held to the same service before costs are compared.**

Sources: paper Fig. 1 (schematic; the yard is generic, not the layout of any real yard).

Earlier work schedules coupled lifts and compares vehicle counts, but never weighs coupling against buying a larger carrier

Shipyard block logistics

- Less empty travel and interference Roh & Cha 2011
- Heterogeneous transporters, delivery limits Kim & Joo 2012; Joo & Kim 2014
- Workload balance, set-ups, precedence, routing Park & Seo 2012
- Paths: distance and number of turns Wang et al. 2020
- Energy and time windows, barge loading, stockyard sequencing Guo et al. 2024; Kweon et al. 2026; Wang & Yin 2026
- More units above 320 t, from fleet utilisation Liu et al. 2022

Cooperative transport

- Vehicles synchronised for one large cargo Hu & Wei 2018
- Multi-vehicle one-cargo for 200–500 t blocks; time and CO_2 Jiang et al. 2021
- Shipyard transporters combined so capacities cover a block; daily selection from a given set Shin & Bak 2014
- One job-shop transport needing several robots Gayon et al. 2025
- Operation, movement and load synchronisation Drexl 2012; Soares et al. 2024

Vehicle numbers and fleet mix

- Shipyard vehicle counts compared; one type, divisible tasks Zhou et al. 2025
- Types and numbers with routes; fixed costs Golden et al. 1984; Hoff et al. 2010; Koç et al. 2016
- AGV and mobile-robot fleets Vis et al. 2005; Vis 2006; Fragapane et al. 2021; Aziez et al. 2022
- Synchronised visits under uncertainty Hashemi Doulabi et al. 2020
- Robot-team composition from simulated schedules Pan & Yu 2024

Analogues, service and search

- Couplable warehouse robots: identical units, only how many Albert et al. 2025; Chaikovskaia et al. 2026
- Modular transit coupling along a corridor Cao et al. 2022
- Fleet size at a stated service level Eskandari & Mahmoodi 2016
- Production scheduling with conflict-free AGV routes Nishi et al. 2011
- ALNS operator performance depends on the problem Voigt 2025

The gap: shipyard work schedules coupled loads and compares vehicle counts; routing and robot-team work link resource choice to performance. Still open: buying single-carrier tonnage or coupling, for indivisible blocks at the same service. We keep teams, alignment, empty travel and due dates in daily scheduling, price fleets from tenders and crew labour, and test how far decisions rest on search and sampled days.

Sources: full references on the last slides (paper §2, Table 2).

Three research questions, and what makes them hard

(i) What to buy §5.1, §5.3

Which fleet sizes and capacity mixes minimise capital and labour costs under one service specification?

(ii) When coupling can compete §3.3, §5.2

Under what workload and cost conditions can cooperative transport compete with single-carrier capacity?

(iii) How service is specified §5.4

How does a lateness cap affect fleet size and composition?

Checks of the findings §5.5

How far do search error, sampled days, parameters and the candidate set move the findings?

Four complications

- **Service**: a fleet can meet an on-time target by letting its hardest blocks slip, so the answer depends on how service is specified
- **Mixes**: light units with a few heavy units offer a middle way between the extremes
- **Insurance**: coupling is often argued for as cover for a broken heavy unit
- **Search error**: counts come from a heuristic, and a rival that qualifies with one transporter fewer can overturn a cost ranking

Sources: paper §1 (research questions (i)–(iii)).

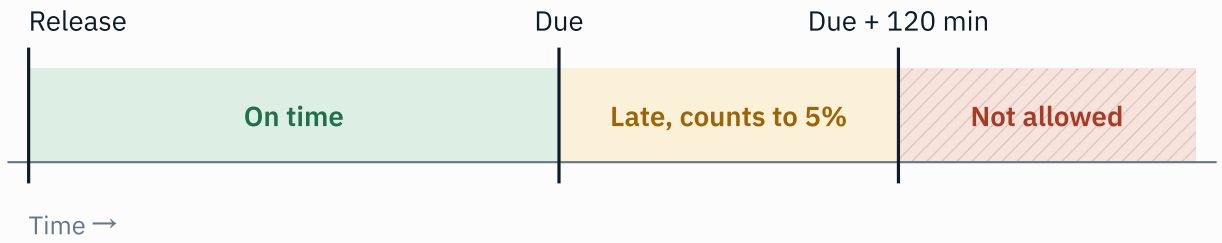
The core idea: hold every candidate fleet to the same service, then compare costs

An analogy

You cannot compare ten buses running 80% on time with twenty minibuses running 99% on time by price alone. First give each fleet the same punctuality, then see which is cheaper.

- For each fleet type, scheduling finds the **smallest count that meets the service, K***, an upper bound on the true minimum
- Then cost K* units: capital plus crew labour per day

≥ 95% on time · T_{max} = 120 min



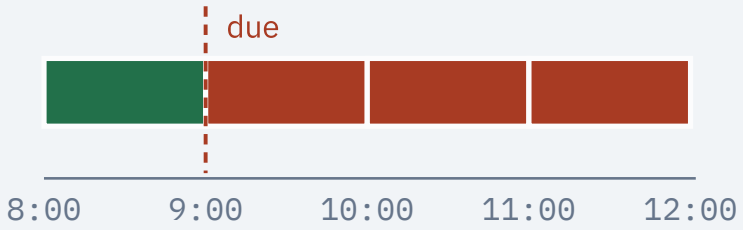
- **95% on time**: at least 2,765 of the 2,910 blocks of 30 days on time (at most 145 late)
- **T_{max} = 120 min**: no block may finish more than 120 min late, one release-batch interval, so a late block cannot overrun the next batch

Sources: the 95% target and T_{max} are our choices: the literature has no common service standard. T_{max} is tested from 30 min to no cap, the target at 90% and 98% (paper §3.1, §5.4).

Why not just divide the total work by one vehicle's hours?

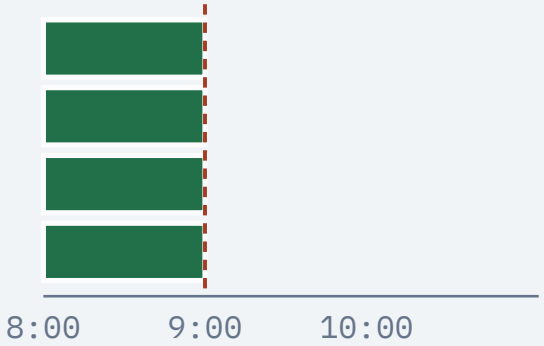
A small example: four blocks ready at 8:00, all due by 9:00, each move takes 1 h

Totals only: 4 vehicle-h ÷ 16 h = 0.25 → one vehicle?



3 blocks late

By due time: four vehicles at once



All on time, but it takes 4 vehicles

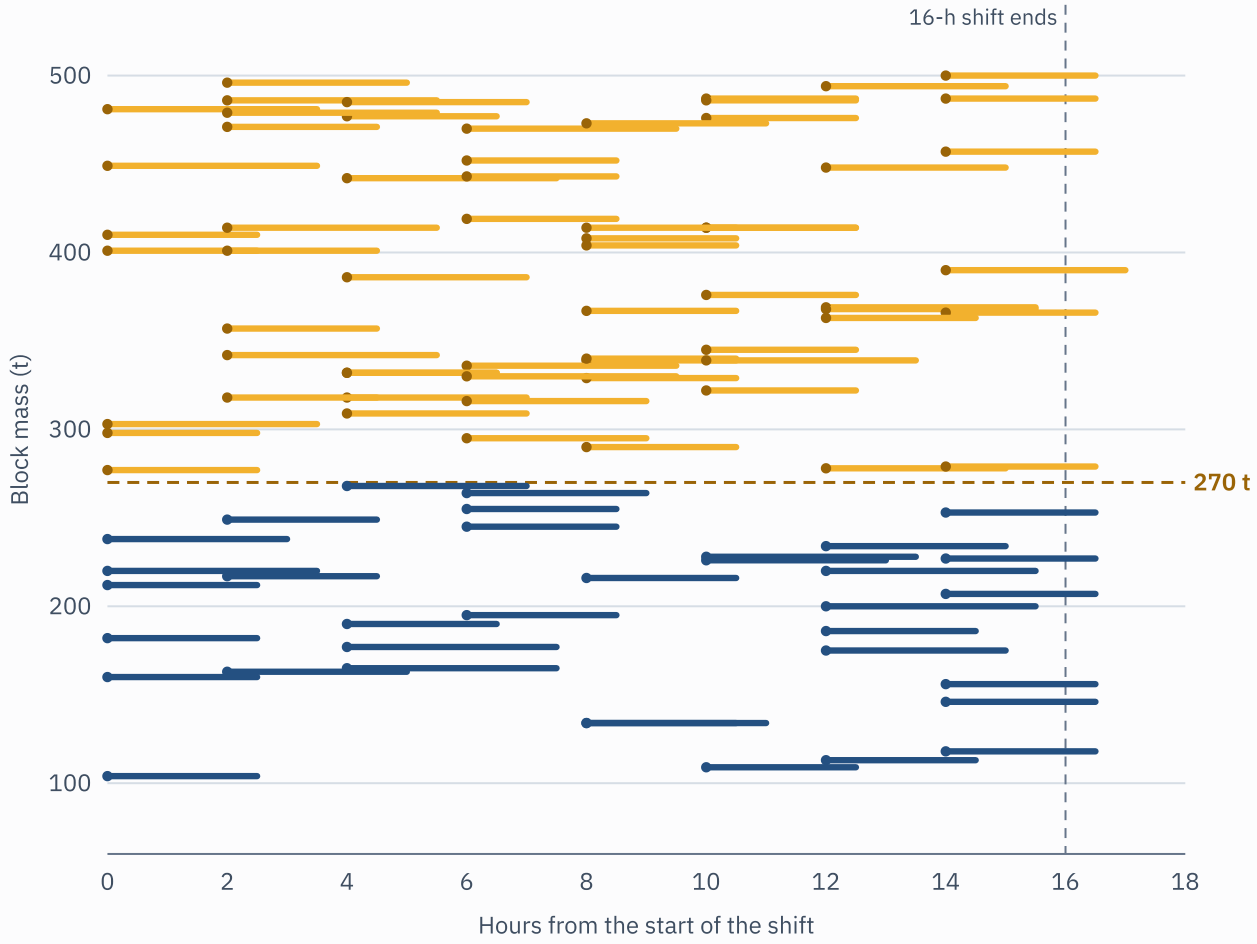
- **Work arrives in batches** with their own due times, so demand is uneven
- **Vehicles drive empty between tasks**, and the order changes that distance
- **A coupled lift needs several free vehicles at once**; one late arrival holds the team

So each day's **actual timetable** must be built. The workload-only count K_ρ is within one transporter of the scheduled count for 319 of 384 sized fleets, but under tight due dates fleets that couple more than 60% of blocks need 10.2% of K* more.

Sources: the example is illustrative; 319 of 384 and 10.2% are results of this study (paper §5.2, Fig. 8c).

One working day: 97 blocks released in eight batches, each with a due time

One generated day (uniform masses · short handling · empirical due dates)



- **97 blocks in 16 h**: the size of a recorded working day (8:00 to midnight)
- **Each line is a block**: left end release, right end due time; height is mass
- **Due times**: 150, 180 or 210 min after release, with probabilities 0.6 / 0.2 / 0.2
- **Release**: a batch every 2 h, about 12 blocks (assumed)
- **Orange** blocks exceed 270 t: a 270 t fleet must couple all of them (62 that day)

Each condition has 30 such days, and every fleet faces exactly the same 30 days.

Sources: 97 blocks and 8:00–24:00 Roh & Cha (2011), p. 3246; due times Roh & Cha (2011) Table 2 (6, 2 and 2 of 10 blocks at 150, 180 and 210 min). Batch release is our assumption.

Who may carry together? Only minimal feasible teams

The team's ratings must add up to the block's mass, and removing any member must make it fall short. Extra units would only waste vehicles and crews.

$$\Sigma\, Q_k \geq m_i \quad \cdot \quad \Sigma\, Q_k - \min Q_k < m_i \quad \cdot \quad |S| \leq 3$$

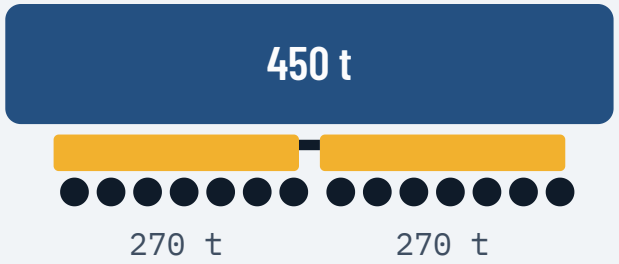
- A block one unit could carry may still go to a light team (flexible coupling, the main rule); the overweight-only rule is compared separately
- For a homogeneous fleet the two rules coincide; only light-plus-heavy mixes differ
- Every minimal team is assumed physically feasible (control, load sharing and road width not modelled), which favours coupling

Try it

Transporter tier (rated capacity)

200 t | 250 t | **270 t** | 300 t | 325 t | 380 t | 425 t | 500 t | 550 t

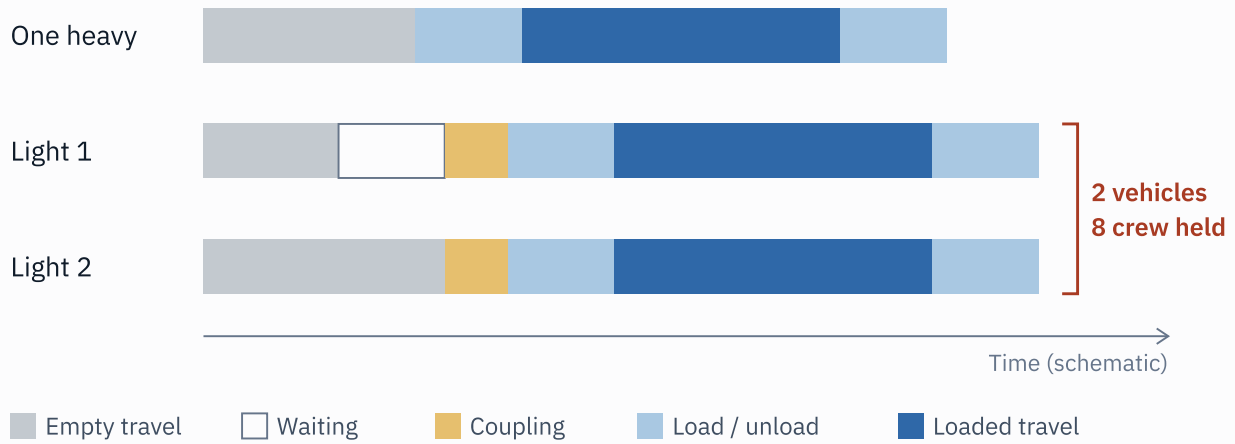
Block mass — 450 t



2 coupled: 2 × 270 = 540 t ≥ 450 t; one fewer gives only 270 t < 450 t. Holds 2 vehicles · 8 crew, plus 10 min coupling.

Sources: big tasks by several flatcars Jiang et al. (2021), constraint (24); overweight-only rule as in Jiang et al. (2021). At most three units is our assumption (two also tested).

The price of coupling: every member must be there, and all are held for the whole move

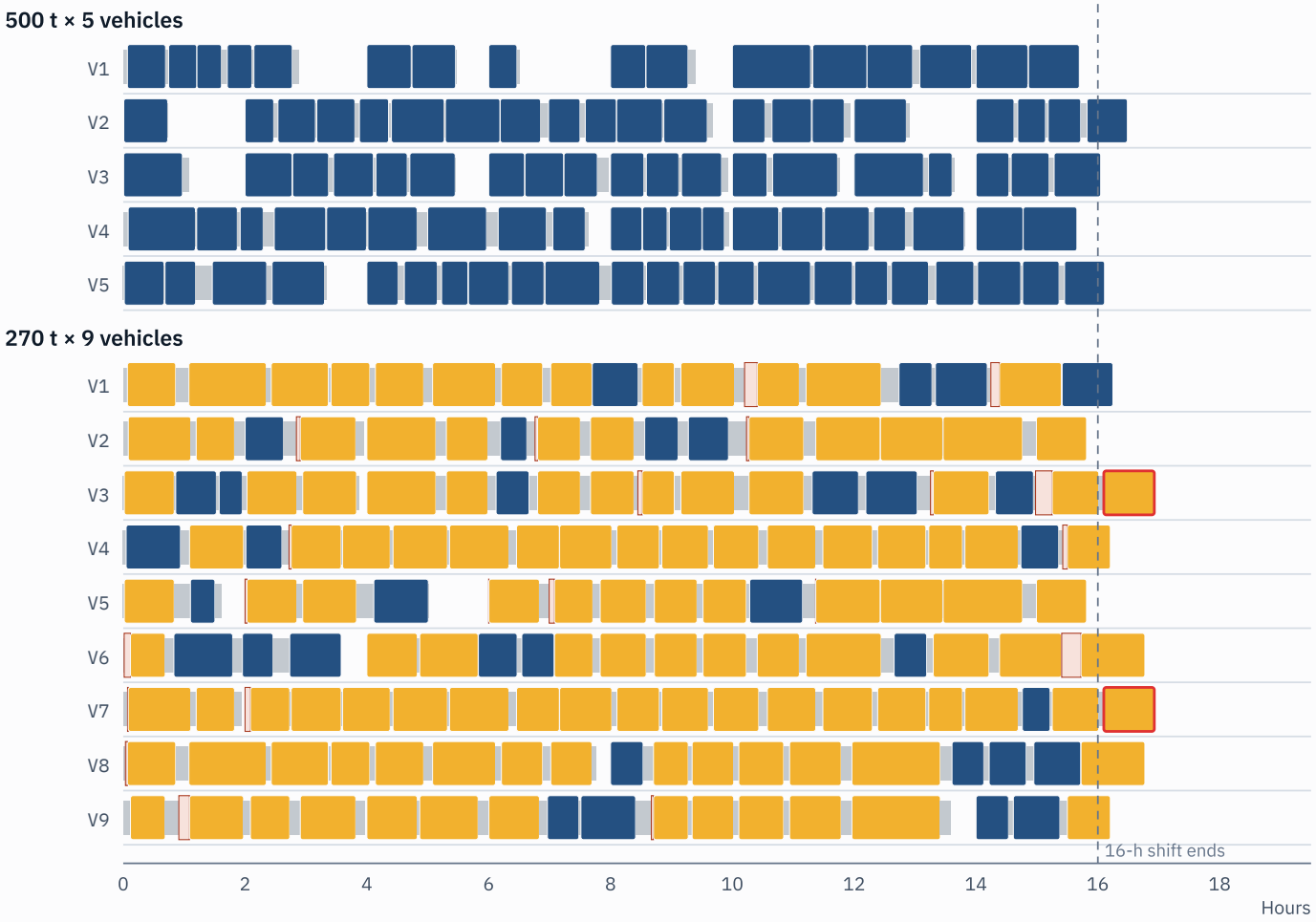


One move up close

- **Alone**: the vehicle loads on arrival and leaves right after unloading
- Example: a 420 t block takes 43 transporter-minutes on one 425 t unit, but 110 on two coupled 270 t units: the second unit's trip 43, alignment 2 × 10 and waiting 4 more
- **Coupled**: the first unit waits for the second (white), then both align (yellow)
- A light block waiting for one of them finishes at 98 min instead of 43

Sources: illustrative times as in Fig. 3 of the paper (43 min alone = 10 empty + 33 service; the 10-min alignment is our assumption).

Same 97 blocks: two fleets' timetables



	500 t × 5	270 t × 9
Vehicles K*	5	9
On time that day	97 / 97	96 / 97
Blocks coupled	0	62
Vehicle work (vehicle-h)	70.0	145.0
· empty travel	5.7	16.7
· waiting for team	0.0	1.9
Worst delay (min)	0	26
Fleet price P (270 t = 1)	8.15	9.00
Daily cost (labour-h, θ = 12.4)	421	688

The 270 t fleet needs nine units and twice the vehicle-hours, coupling 62 blocks; here even its capital is higher.

Rows are vehicles, bars tasks (hover for the block); stored schedules at each fleet's smallest qualifying count, re-simulated.

Sources: schedules of this study (uniform masses · short handling · empirical due dates, first day).

How timetables are found: far too many to try, so we search cleverly

$$97! \approx 10^{152}$$

The order of 97 blocks alone has about 10^{152} possibilities, before choosing any teams.

So we use **adaptive large neighbourhood search (ALNS)**, improving a workable timetable step by step.

1 Destroy

Remove some blocks: random, worst, related, a team or a route (5 ways); 2 more switch single ↔ coupled

→

2 Repair

Reinsert them (greedy or regret-2), choosing a vehicle or team for each

→

3 Accept

Keep the new timetable if it is better; now and then accept a slightly worse one so the search does not get stuck

→

4 Learn

Record which destroy and repair moves pay off and use them more often: the "adaptive" part

↻ 786 rounds per run (38 s on average), exactly repeatable from its seed; the start is the best of 5,000 random chronological constructions

How two timetables are ranked

① blocks over 120 min late → ② late blocks → ③ total lateness → ④ crew time. One more block on an earlier item outweighs any gain on later ones. A team's members are updated together at every step, so coupled lifts stay synchronised.

Sources: ALNS framework and adaptive weights Ropke & Pisinger (2006), Pisinger & Ropke (2007); operator types Voigt (2025). Team dissolution, the two mode operators and the choice of a team for each inserted block are specific to coupled lifts. Iteration and construction counts are our decisions; search depth is checked with counts after constructions only, 100 and 300 rounds (paper §3.4, Supplementary S1).

Capital and labour on one scale: labour-hours

Vehicle

CNY 2.21 M

Price of one 270 t transporter (ex VAT, fitted to award prices)

→

CNY 329 k / yr

Spread over 8 years at 4% interest (capital recovery factor 0.1485)

→

1,316 CNY / day

250 working days a year

Labour

79,182 CNY / yr

Average wage of manufacturing production workers in China (NBS, 2025)

→

× 1.3 ÷ 2,000 h

Plus 30% employer on-costs (social insurance etc.), 2,000 hours a year

→

51.5 CNY / labour-h

Cost of one worker for one hour

Price ratio θ

θ = 1,316 ÷ 51.5 = 25.6

One 270 t transporter's daily capital cost = 25.6 labour-hours

Compare

64

One transporter's own crew (a driver and three guiders) per day: 4 × 16 h = 64 labour-hours

Total cost

C = θ·P + L

P: the fleet's price in units of one 270 t transporter; L: the fleet's labour-hours per day

With Shanghai manufacturing wages (CNY 120.1 per labour-hour) θ = 11.0. Over the combinations of life, interest, working days, on-cost and the two wages, θ lies in 5.1–30, and every result is reported over that range.

Sources: 270 t price fitted to 33 public tender awards (ex VAT); wages National Bureau of Statistics of China, release of 15 May 2026, 2025 average wages, Table 7 (large enterprises · manufacturing · production workers, CNY 79,182), and China Statistical Yearbook 2025, Table 4-12 (Shanghai manufacturing, CNY 184,833); on-cost from official contribution rates totalling ≤ 39.4% (Shanghai HRSS 2025; State Council 2019 No. 13; MOHRSS 2015 No. 71; MOHURD 2016 No. 74); crew of four, the lower end of the four or five per transporter in practice, Park & Seo (2012), p. 392. Service life, interest, working days and 2,000 h a year are our decisions; θ is also tested over 0.5–5,000 (paper §3.2, Table 3).

Four ways to count crew labour; 180 cost scenarios per condition, 6,480 decisions in all

Shift labour

64 · K

Every transporter staffed by a crew of four for the 16-h day. The primary measure: it is the staffing a yard must provide, and it grows with K.

Shift + overtime

64·K + after-shift

Adds 1.5 × 4 crew-hours for every transporter-hour worked after the shift.

Crew-vehicle hours

4 × vehicle-hours

Four crew for the hours each transporter actually works (empty travel, waiting and service).

Crew-team hours

4 per team when coupled

Four per vehicle for empty travel and waiting, but only four per team during a coupled lift, as if one crew could drive the set: the most favourable to coupling.

`5 price curves × 9 values of θ × 4 labour measures = 180 cost scenarios per condition · × 36 = 6,480 decisions`

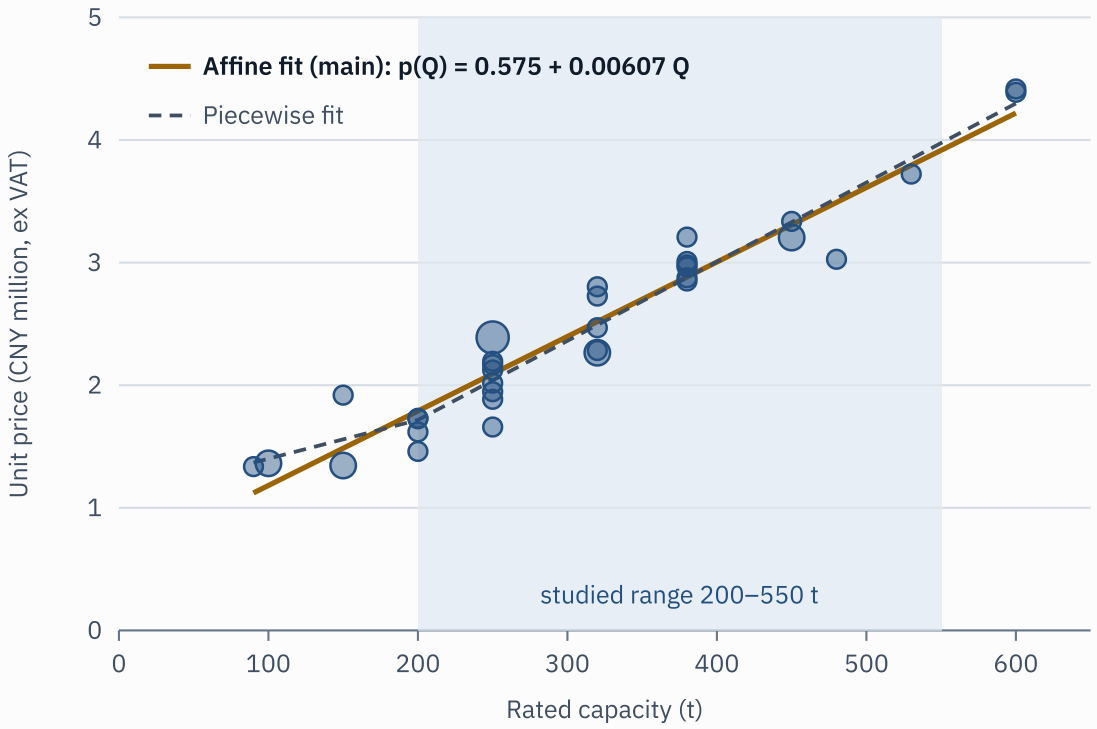
- In each scenario the fleet with the lowest C_X is the **least-cost fleet**; the runner-up's excess over it is the **margin**
- The last three measures re-price **the same schedules**; schedules are not re-optimised for each
- **Crews**: every transporter has its own crew of four (one driver, three guiders), not shared; loading and unloading count as crew time, and shop or stockyard staff outside the crews are not counted

The 6,480 decisions re-price the same 384 sized fleets under different cost parameters; they are **not independent experiments**, and shares of decisions are descriptive frequencies over the grid, not probabilities.

Sources: crew of four, the lower end of the four or five per transporter in practice, Park & Seo (2012), p. 392; the overtime factor 1.5 is our setting; measures defined in paper §3.2, scenarios and decisions in §3.4.

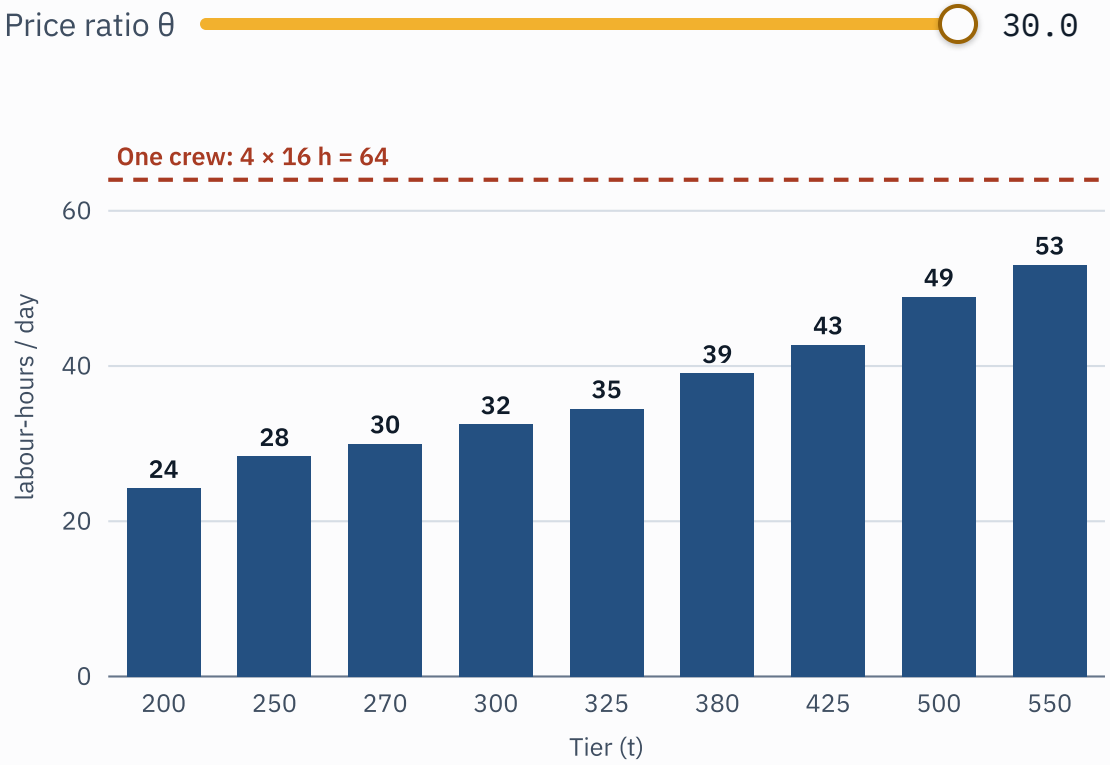
Prices: a vehicle-day of capital costs less than a crew-day

Unit prices of 39 transporters (33 public award notices, 2022–2026, ex VAT); shaded: studied range 200–550 t



The 27 records within 200–550 t give α = 0.84 (95% CI 0.72–0.96); over all 33 records a lack-of-fit diagnostic (p = 0.039) argues against a single power law, so five price curves are used: affine (main), piecewise and α = 0.72, 0.84, 0.96.

Per transporter-day: capital (bars) against one crew (dashed), in labour-hours

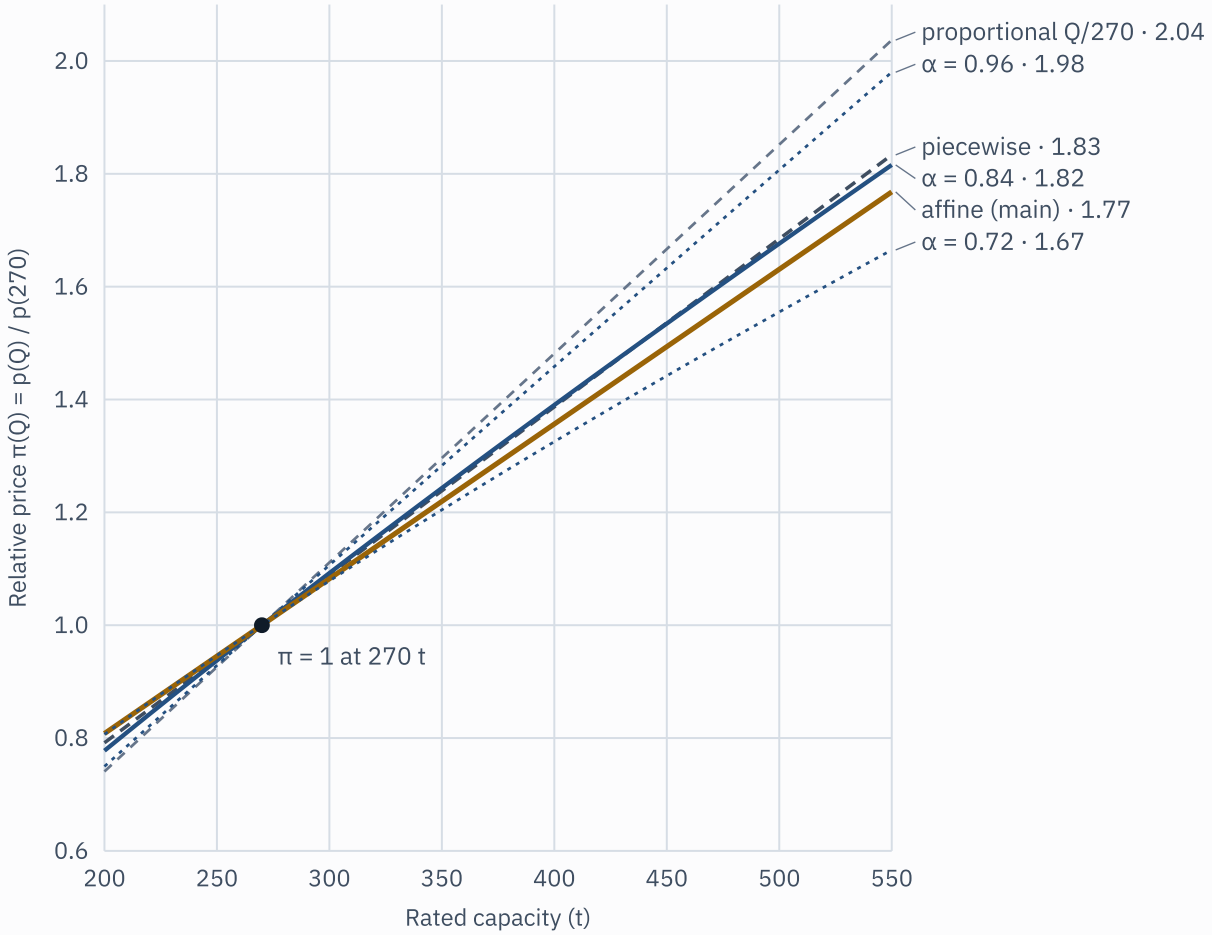


At θ = 30.0, the dearest tier, 550 t, costs 53 labour-hours of capital a day, still below one crew’s 64. Saving a transporter beats moving up a tier.

Sources: 33 award notices from csscbidding.com, China Shipbuilding Electronic Bidding Platform (26), 365trade.com.cn (5) and e.sinochemitc.com (2), each listed in Supplementary Table S4; fits and tests are our computation (paper §3.2, Table 3, Fig. 2a).

Five price curves: above 270 t, every curve lies below proportional pricing

Price relative to a 270 t transporter, π(Q) = p(Q) / p(270), over 200–550 t; numbers on the right are the values at 550 t



- **Affine** (main): p(Q) = CNY 0.575 million + 6,075 per t; **piecewise**: CNY 1.085 million + 3,159 per t, plus 3,295 per t above 200 t. Both are fitted to all 33 notices
- **Power laws** $p \propto Q^{\alpha}$: the 27 notices within 200–550 t give α = 0.84 (95% CI 0.72–0.96; bootstrap over 20 buyers 0.73–0.95); 0.72 and 0.96 bound the interval
- One power law fits all 33 notices poorly (lack of fit F(9, 22) = 2.49, p = 0.039); within 200–550 t no lack of fit is detected (F(5, 20) = 0.86, p = 0.52)

> **One large unit costs no more than two smaller ones together (sub-additivity): the property on which single-carrier dominance rests.**

Sources: curves fitted to the 33 award notices (ex VAT, Supplementary Table S4); our computation (paper §3.2, Table 3, Fig. 2b).

Single-carrier dominance: for every block, one large-enough transporter costs less than any coupled team

One 420 t block (affine prices, θ = 12.4, shift labour, short handling)

	ONE 425 T ALONE	TWO 270 T COUPLED
Capital π	1.43	1 + 1 = 2.00
Crews	1	2
Occupancy	43 min	2 × (43 + 10) min
Cost per block	**3.7 labour-h**	**8.4 labour-h**

$$\text{cost per block} = (w_i + \delta \cdot 1[|S| > 1]) \div H \times \Sigma_{k \in S}\, \gamma(Q_k),$$
$$\gamma(Q) = \theta \cdot \pi(Q) + \ell$$

Fluid approximation: transporters divisible and fully usable over the day, counting only the vehicle-time each block takes; due dates, integer counts and schedule geometry are ignored.

Proposition 1: with sub-additive prices (π(a + b) ≤ π(a) + π(b); all five price curves over 200–1,650 t), if the menu has a tier Q* with $m_i \le Q^* \le$ the team's capacity, one transporter of Q* carries block i at a strictly lower fluid cost than any minimal team of g ≥ 2.

- **No more capital**: one larger unit costs no more than the members together
- **g – 1 fewer crews**: crews are paid per vehicle
- **No alignment**: every member is held δ longer

It holds with labour alone (θ = 0) and with capital alone (ℓ = 0, δ > 0); even at δ = 0 a coupled lift adds (|S| – 1)·w + |S|·δ of transporter-time.

> **Corollary 1: if every block has a just-large-enough tier, the fluid-optimal fleet couples nothing and is mass-matched.**

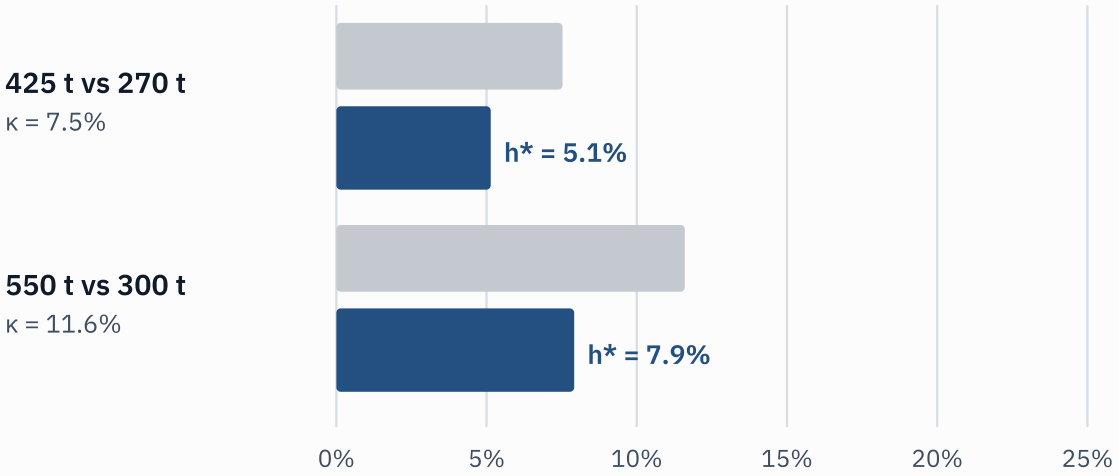
Sources: prices: five curves fitted to 33 tender awards; 43 min: the covering fleets' mean occupancy per block with short handling (this study); 10-min alignment assumed. Paper §3.3, Assumption A, Proposition 1, Corollary 1.

The coupling budget: how much of its workload can a light tier couple and still beat the covering tier?

Price ratio θ 12.4

Alignment δ (min) 10.0

Occupancy w̄ (min) 43



κ: how much dearer the larger unit is per day

h*: the largest coupled share of workload the light tier can afford

h* = κ ÷ (1 + 2δ / w̄) = κ ÷ 1.47; δ = 10.0 min, w̄ = 43 min (covering fleet's mean occupancy per block: about 43 with short, 63–85 with mass-dependent handling, about 80 with long loading); α = 0.84 power law, shift labour

$$h \cdot (1 + 2\delta / \bar{w}) < \kappa \Leftrightarrow h < h^{*} = \kappa / (1 + 2\delta / \bar{w})$$

- **h**: the coupled share of single-carrier occupancy (workload, not count); **κ** = $\gamma(Q_{hi}) / \gamma(Q_{lo}) - 1$, how much dearer the covering tier is per day
- Labour dilutes the capacity premium: for θ = 5.1–30, κ is only 3.4–14.8% (425 vs 270 t) and 5.3–22.5% (550 vs 300 t)
- Covering fleets occupy a transporter about 43 min per block with short handling, 63–85 with mass-dependent handling and about 80 with long loading; with δ = 10 min the factor is 1.47 or 1.24–1.32
- So **h* ≈ 2–18%** under shift labour and the fitted power law (1.9–21.8% across the five price curves); yet in the 36 conditions the 200–380 t tiers couple **5–100%** of blocks

The budget is small because labour dilutes the capacity premium.

Sources: κ from the power law fitted to 200–550 t tender prices (α = 0.84) with shift labour, θ = 5.1–30; occupancy per block and 5–100%: this study (paper §5.1, §5.2); 10-min alignment assumed. Paper §3.3, Corollary 2.

What the model leaves out, and the four implications the results test

In real operations but not in the fluid model (each can favour coupling)

Menu gap

If no tier can carry a block, coupling is necessary.

Integrality

Transporters are indivisible: a heavy tier kept for a thin heavy tail may stand idle; a light fleet with one or two heavy units is the integer counterpart of a mass-matched fleet.

Timing

Release peaks and due dates need slack beyond average workload, and a coupled lift needs several vehicles free at once.

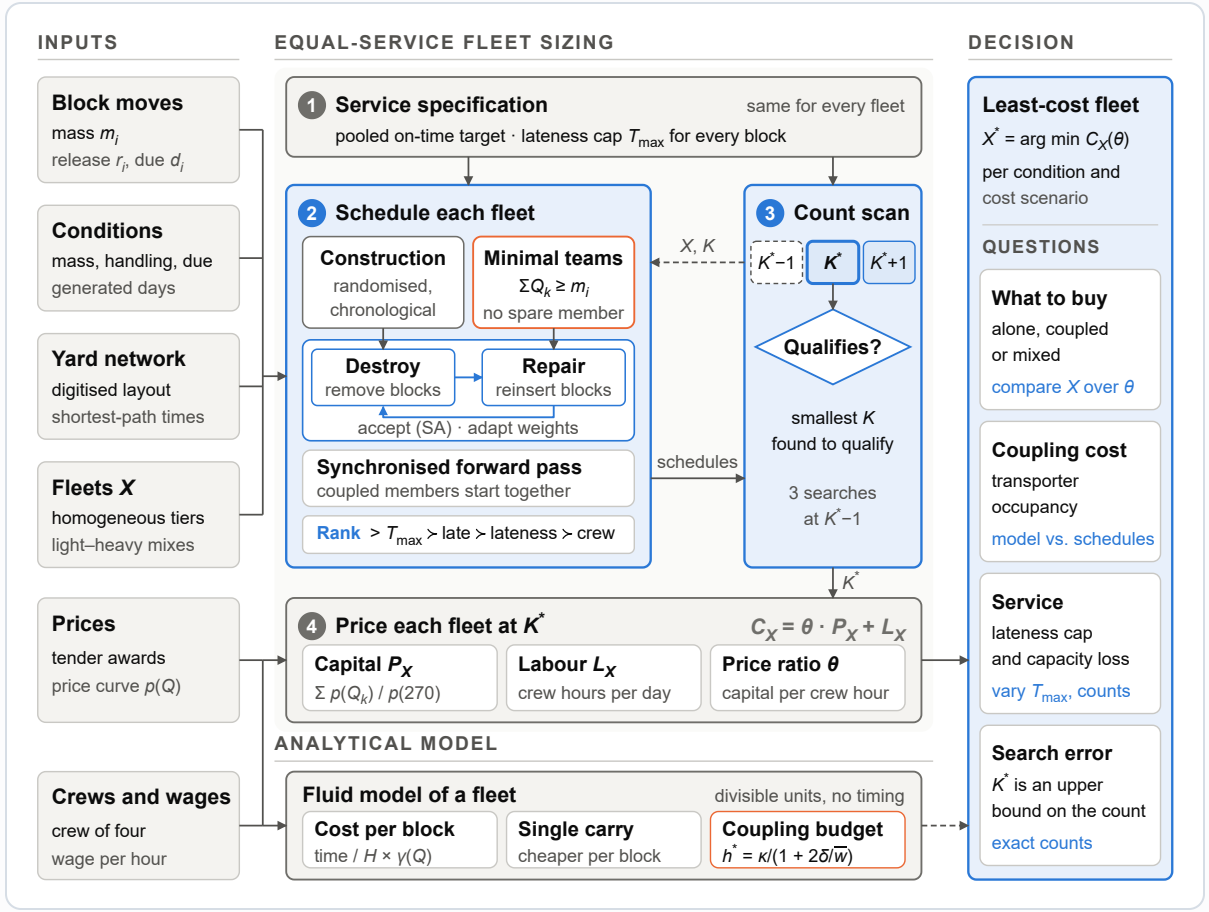
Four implications

1. Where some tier carries every block, least-cost fleets couple little
   → Results: what to buy
2. A light tier beats the covering tier only when its coupled occupancy is below h*
   → Why coupling costs: the calibrated screen
3. The heavy units of the cheapest light–heavy mix track the heavy blocks' share of occupancy
   → Composition: the heavy-share rule
4. Coupling-reliant fleets can win only where the budget does not bind or is large: a menu gap, near-zero alignment, or capital dear relative to labour
   → Capital against labour; when coupling pays

Two kinds of heavy: blocks above a light tier but not the largest can be served by buying capacity and govern the purchase; blocks above every tier must be coupled whatever the fleet and only set a floor on coupling.

Sources: paper §3.3.

Evaluation procedure: specify service, schedule each fleet, scan its count, price it at K*



- ① **Service specification**, the same for every fleet: ≥ 95% on time pooled over 30 days, and every block within T_{max}
- ② **Schedule**: ALNS with single and coupled lifts; rank by over T_{max} ≻ late ≻ lateness ≻ crew time
- ③ **Count scan**: the smallest K found to qualify; three searches a day at K* − 1; a schedule for K stays feasible for K + 1
- ④ **Price**: $C_X = \theta \cdot P_X + L_X$; the least-cost fleet in each cost scenario

Search error enters at step ③, where K* is an upper bound, and can change the decision at step ④; the checks of the findings test both. All fleets face the same blocks, locations and start points (common random numbers); a fleet of K uses the first K of 40 start points.

Sources: evaluation procedure of this study (paper §3.4, Fig. 4), drawn in the colours of the paper's figures.

Case: 36 conditions on one digitised yard, plus a second yard rebuilt from published data

Eight flows and their relative weights (assumed)

Assembly shop → Outfitting shop	0.5
Assembly shop → Paint shop	0.5
Outfitting shop → Paint shop	0.5
Paint shop → Block yard	0.7
Paint shop → Assembly yard	0.3
Block yard → Assembly yard	0.7
Assembly yard → Dock	0.2
Assembly shop → Assembly yard	0.05

The road network is the digitised layout of a large shipyard (not released); travel times are static shortest paths at 12 km/h empty and 6 km/h loaded (270 t data sheet), without traffic interference; pickup and drop-off points are drawn within the zones; eight batches a day, one every 2 h; each transporter starts at a random shop entrance. Process routes (the sequence of shops, stockyard and dock) and road paths are inputs, not optimised, so every candidate fleet serves the same demand.

36 × 30

6 mass profiles × 3 handling regimes × 2 due-date regimes; 30 days each: 1,080 days, 104,760 block moves

11

reference fleet types: 9 tiers (200–550 t) and 270 t units with one or two heavy units (scheduled as 550 t, priced at the smallest tier that carries every block they carry alone)

384 · 6,480 sized fleets (200 and 250 t not run for the heavy tail) · decisions

214,210

scheduling runs (51,480 in the main experiment), about 2,570 core-hours (Intel Core Ultra 9 275HX, 24 cores, 32 GB RAM, Windows 11, PyPy 3.9); each seeded and budgeted by iterations, so it repeats exactly

50

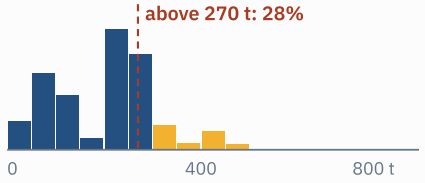
second yard: the published week of transfers, distance table and five units (250–420 t) of a Shanghai cruise-ship yard

Sources: 97 blocks a day Roh & Cha (2011); speeds Suzhou Dafang DCY270 data sheet; flow categories and weights are assumed; second-yard data Liu et al. (2022), Tables 4–5 and supplementary data S1; run counts are ours (paper §3, §4.1, §4.3).

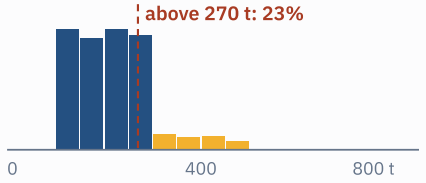
36 conditions = 6 mass profiles × 3 handling regimes × 2 due-date regimes

Six mass profiles: masses of the 2,910 blocks of 30 days (orange: above 270 t)

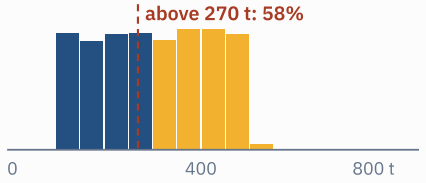
Jiang · 38–450 t
68 masses from the three instances of Jiang et al. (2021), resampled



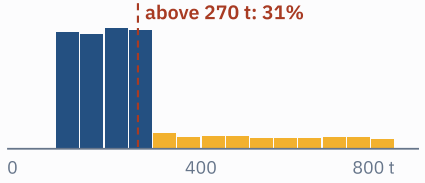
Light-skewed · 100–500 t
90% at 100–300 t (Park & Seo 2012), 10% at 300–500 t (assumed)



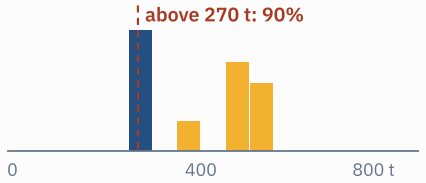
Uniform · 100–500 t
Uniform 100–500 t (the rule of the public instances of Kweon et al. 2026)



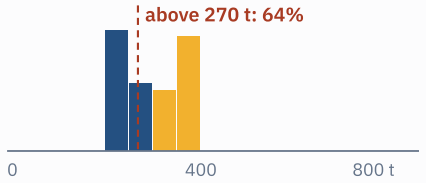
Heavy tail · 100–800 t
Light-skewed plus a 10% tail at 500–800 t: some blocks exceed every tier (assumed, a stress test)



Roh–Cha · 260–544 t
The ten block masses of Roh & Cha (2011) Table 2, resampled



Liu · 204–399 t
The 50 block masses of one week in Liu et al. (2022), resampled



Three handling regimes

- **Short**: 10 min each (Liu et al., 2022)
- **Mass-dependent**: U(10, 20) + 0.05 × mass − 5 min each (Kim & Joo, 2012)
- **Long loading**: 30–60 min loading, 10 min unloading (Liu et al., 2022)

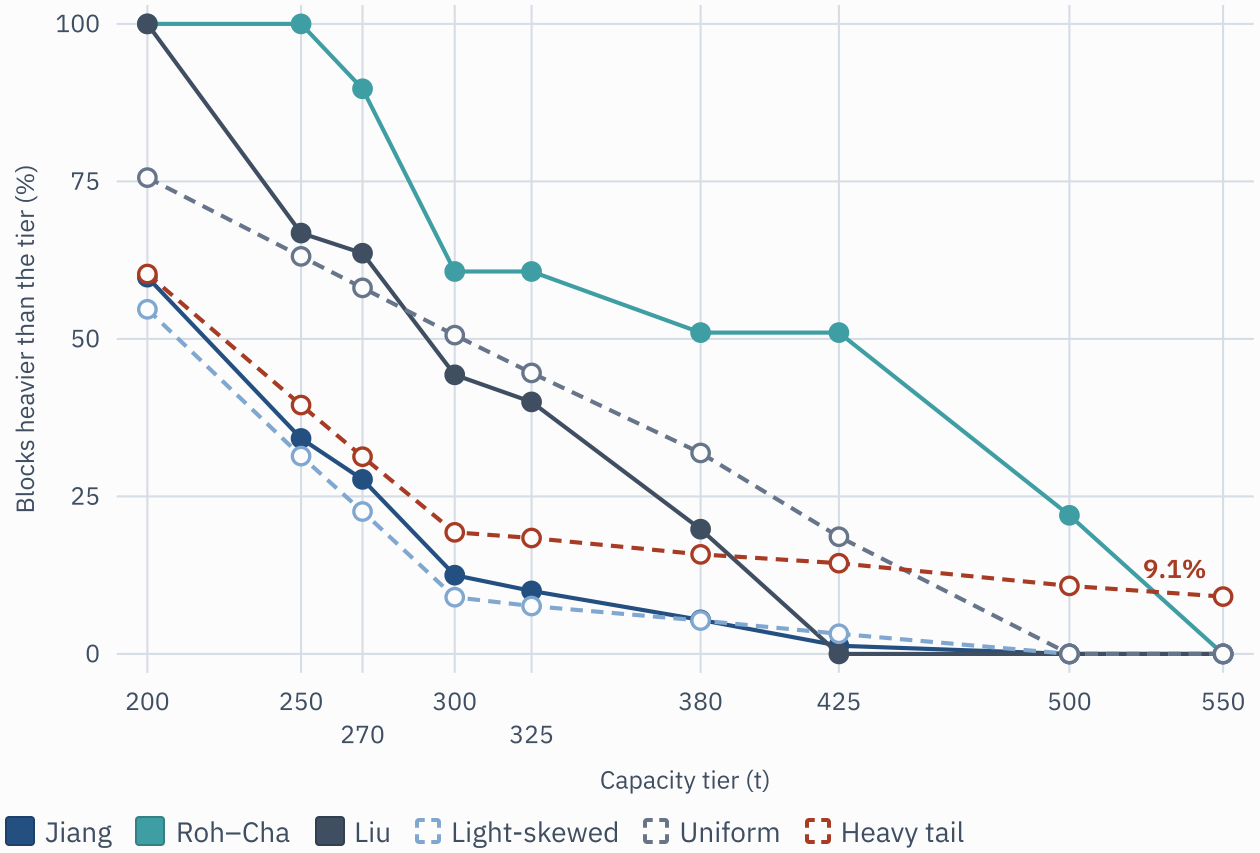
Two due-date regimes

- **Empirical**: 150 / 180 / 210 min with probabilities 0.6 / 0.2 / 0.2 (Roh & Cha, 2011)
- **Tight**: 120–150 min (assumed, tighter than the shortest published 150 min)

Sources: masses Jiang et al. (2021) Tables 5, A1, A2 (68 after removing repeats); Park & Seo (2012), p. 390 (100–300 t); Kweon et al. (2026) public instances (100–500 t); Roh & Cha (2011) Table 2; Liu et al. (2022) S1 data. The 10% at 300–500 t in the light-skewed scenario and the 500–800 t tail are assumptions. Handling Liu et al. (2022) §4.1.1, §4.2.2; Kim & Joo (2012) §4. Due dates Roh & Cha (2011) Table 2. Paper §4.1, Table 4, Fig. 5a.

Blocks heavier than each tier: the share a homogeneous fleet of that tier must couple

Share of blocks heavier than each capacity tier (2,910 blocks per profile; solid: published samples, dashed: stylised profiles)



- **Covering tier** (the smallest tier that carries every block alone): 500 t for Jiang, light-skewed and uniform, 550 t for Roh–Cha, 425 t for Liu
- The **heavy tail** has none: 9.1% of blocks exceed 550 t and must be coupled by any fleet
- Only 1.3% of the Jiang blocks exceed 425 t; 44% of the Liu blocks exceed 300 t
- So across the 36 conditions the 200–380 t tiers couple 5–100% of blocks

Below the covering tier the masses, not the schedule, fix how much a fleet couples.

Sources: the 2,910 block masses of each profile in the main experiment (released with the paper); this study (paper §4.1, Fig. 5b).

Parameter sources (1): values taken from the literature or public data

PARAMETER	VALUE	SOURCE (PAGE / TABLE)
Moves per day · shift	97 · 16 h	Roh & Cha (2011), p. 3246
Baseline due times	150 / 180 / 210 min (0.6 / 0.2 / 0.2)	Roh & Cha (2011) Table 2
Mass samples	38–450 t · 260–544 t · 204–399 t	Jiang et al. (2021) Tables 5/A1/A2; Roh & Cha (2011) Table 2; Liu et al. (2022) supplementary data
Uniform masses · footprint	100–500 t · 5–15 m by 10–30 m	Kweon et al. (2026) public instances
Handling	10 min each · mass-dependent · 30–60 min loading	Liu et al. (2022) §4.1.1, §4.2.2; Kim & Joo (2012) §4
Speed	12 empty · 6 loaded km/h; by model 10/5, 12/5	Suzhou Dafang DCY270, DCY380, DCY500 data sheets
Tiers	200–550 t	Jiang et al. (2021) Table 5; Roh & Cha (2011) Table 1; Liu et al. (2022) Table 4
Crew per vehicle	four: one driver and three guiders, not shared (published: four or five; lower end)	Park & Seo (2012), p. 392
Prices	39 units, CNY 1.34–4.42 M, 90–600 t	33 award notices: csscbidding.com (26), 365trade.com.cn (5), e.sinochemitc.com (2), 2022–2026 (Supplementary Table S4)
Wages	CNY 79,182 a year (national, 2025) · 184,833 (Shanghai, 2024)	National Bureau of Statistics of China (2026), Table 7; China Statistical Yearbook 2025, table 4-12
On-cost	30–40%	official contribution rates, ≤ 39.4% in total

Sources: each entry was checked against the original page and sentence; full references on the last slide.

Parameter sources (2): values we had to set, and how each was tested

PARAMETER	SETTING	TESTED AT
Release	eight batches, 2 h apart	identical for all fleets
Tight due dates	U(120, 150) min	a condition beside the published due dates
Heavy part of light-skewed · heavy tail	10% × 300–500 t · 10% × 500–800 t	stress conditions; the heavy tail is analysed separately
Alignment δ	10 min	0, 2.5, 5, 20, 30, 40 min: low-coupling share 88.6% at δ = 0, 100% at 20–40 min
Team size · turns	≤ 3 · 0 s	at most two; 30 s per turn of 60° or more; 30 or 60 s per turn for coupled teams
One speed for all tiers	12 / 6 km/h	manufacturer speeds 10/5, 12/5 km/h; speed × 0.75, × 0.5; Liu's 50/30 m/min
Handling multiple	× 1	× 0.5, 0.67, 1.5, 2; plus ±50% random noise
Service	95% · 120 min	caps of 30, 60, 240, 480 min and none; on-time targets of 90% and 98%
Capital recovery · hours	8–15 y, 4–8%, 250–300 days; 2,000 h a year	θ reported over all of 5.1–30, then 0.5–5,000
Sampled days	30 per condition	2,000 bootstrap resamples; 30 fresh days
Search budget	786 rounds; 5,000 constructions	counts after constructions, 100 and 300 rounds; six more searches a day at K* − 1; exact counts on small instances

Sources: results of these tests are in the composition-and-service and validation parts (paper §5.3–5.5, Supplementary S2–S5); manufacturer speeds Suzhou Dafang DCY270, DCY380, DCY500 data sheets; 50/30 m/min Liu et al. (2022) §4.1.1.

Confirmation experiments: the model's implications tested on new instances, and an exact benchmark

Fresh days
30 days never used elsewhere; the 42 fleets contending for least cost in seven conditions

Extended mixes
13 further conditions: 300 t units with $K_H^* - 2$ to $K_H^* + 2$ covering heavy units (500, 550 or 425 t)

Slow speed and long handling
425 t, 500 t and mixes added at slow speeds and at handling ×1.5 and ×2; unresolved fleets scanned up to 40

Capacity loss
one fewer transporter in each capacity class, start positions reassigned; redundancy estimated from qualifying counts

On-time target
the 42 fleets re-sized at 90% and 98%

Exact benchmark
80 single-batch slices of 12–13 blocks from ten days in eight conditions; CP-SAT computes the fewest late blocks at each count; 70 sized fleets

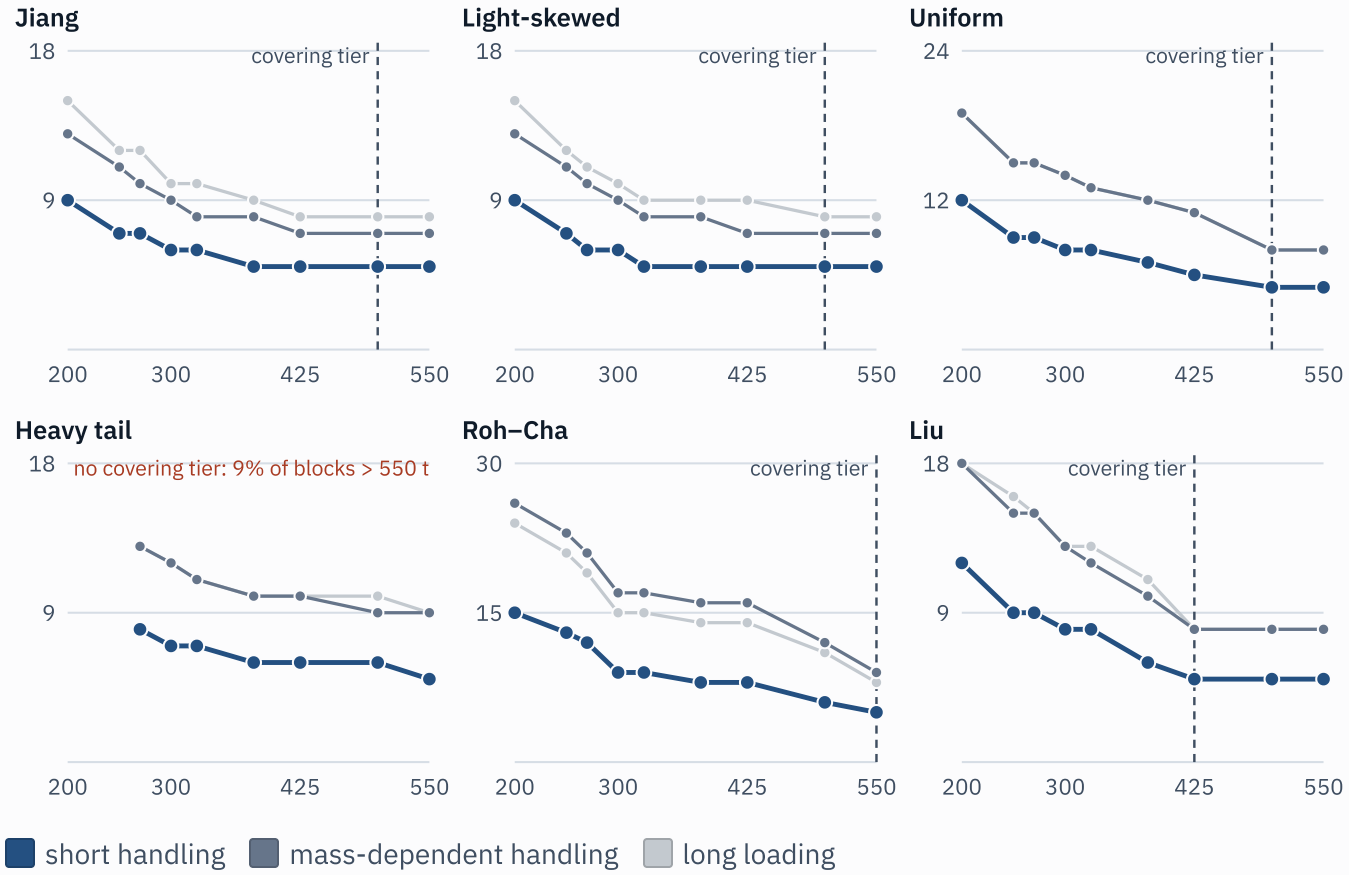
Heavy-share rule: $K_H^* = \lceil K_{cov} \cdot h_H \rceil$

K_{cov}: the count of the covering homogeneous tier; h_H: the share of loading and unloading time taken by blocks above 300 t. Three further analyses use the stored runs: a redundancy estimate from the qualifying counts of reduced compositions, schedule re-selection for each labour measure, and a block-by-block decomposition of transporter-time.

Sources: paper §4.2, Table 4; Supplementary S5, S7. Exact solves use CP-SAT of Google OR-Tools.

Question (i): larger tiers need fewer transporters until one tier carries every block alone; beyond it, no further saving

Smallest qualifying count K* (empirical due dates); dashed: the covering tier, the smallest tier that carries every block alone



- **Liu masses**, short handling: 12 units of 200 t, 9 of 270 t, 6 of 380 t, 5 of 425 t or more
- **Roh–Cha masses**: 15, 12, 8 and 5 units of 200, 270, 425 and 550 t
- Slower handling raises every line (up to 26 units of 200 t under Roh–Cha masses, 30 with tight due dates); the shape stays
- Beyond the covering tier extra capacity buys nothing: under Liu masses 425, 500 and 550 t need the same count in all six conditions
- Below it, the masses fix the coupled share: the 200–380 t tiers couple 5–100% of blocks

A light fleet's extra units are the price of coupling.

Sources: this study: count scans of 11 fleet types in 36 conditions (paper §5.1, Fig. 6).

Question (i): the least-cost fleet carries almost every block alone

91.0%

of 1,620 shift-labour decisions: the least-cost fleet couples ≤ 1/10 (all 1,350 where a covering tier exists)

57.5%

of all 6,480 decisions: the winner couples no block

24.1%

won by light units with one or two heavy units, coupling ≤ 1/10: each carries its own blocks

1.2%

won by a light tier (≤ 325 t), only in the heavy tail or the hardest Jiang condition

By labour measure (share coupling ≤ 1/10, 1,620 decisions each)



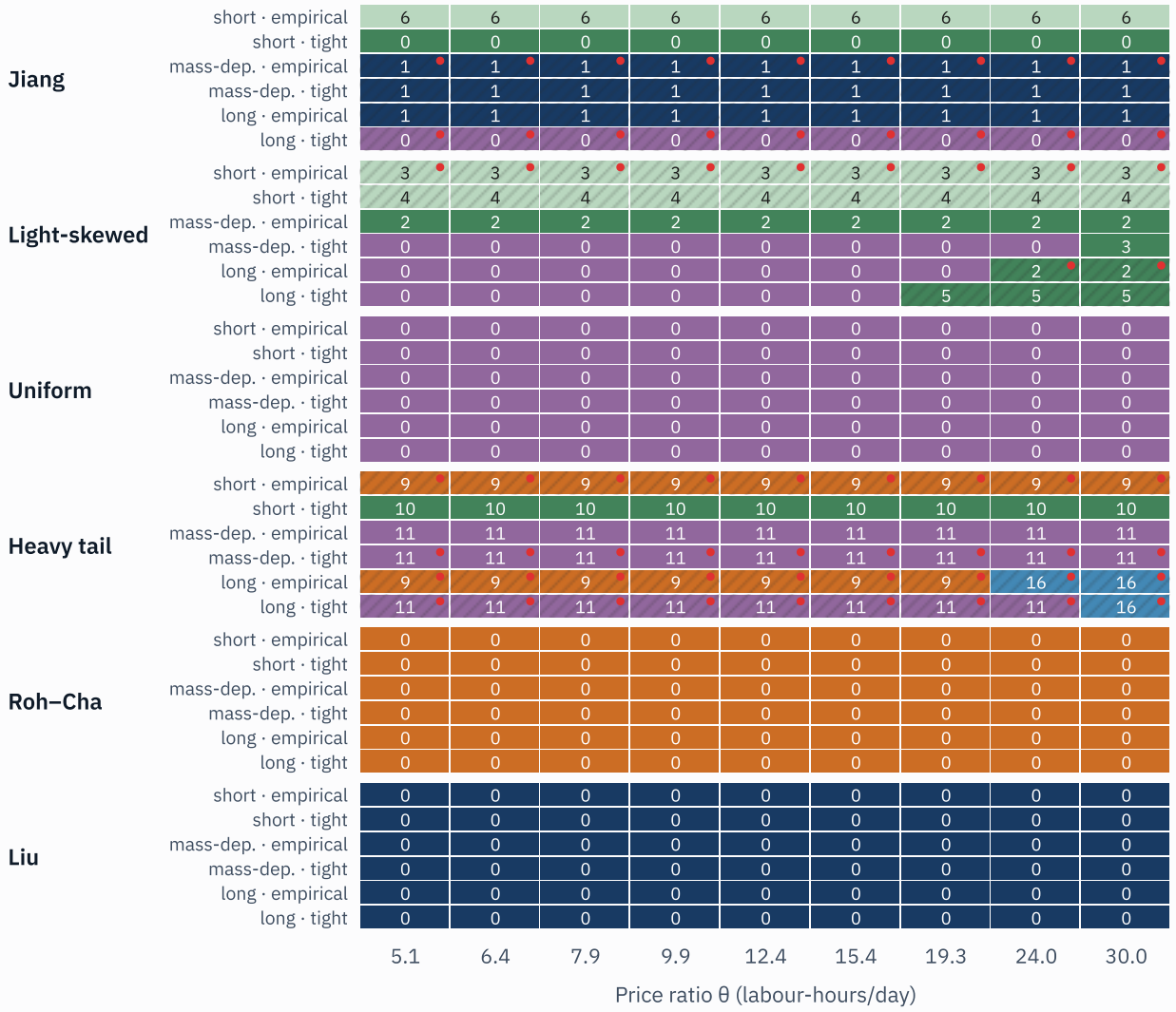
* one crew drives a coupled team, the measure most favourable to coupling; outside the heavy tail, all 32 decisions won by a coupling-reliant fleet use it.

- Over all four measures 90.8% (5,882 of 6,480); homogeneous fleets carrying alone win 66.7%
- **The threshold is not critical**: ≤ 2.5% in 74.5%, ≤ 15% in 97.3%; price curves 90.0–91.6%; θ 5.1 → 30: 93.1% → 87.6%

> **Every minimal team is assumed physically feasible, which favours coupling, so that assumption does not explain these shares.**

Sources: this study: 6,480 decisions (paper §5.1, Fig. 7); a decision = condition × price curve × θ × labour measure (36 × 5 × 9 × 4).

All 36 conditions × 9 price ratios: which fleet is cheapest



Shift labour | Crew-team hours

- **Colour**: the winning fleet; **number**: the share of blocks it couples (%)
- **Hatched**: winner kept in < 80% of day resamples (a near tie)
- **Red dot**: a decision that more search might flip

380 t · 425 t · 500 t · 550 t · 270 t + 1 heavy · 270 t + 2 heavy
support < 80% · weak search dependence

Example: the bottom six rows (Liu masses) are all dark blue, so 425 t is cheapest at every price, and the 0 means it never couples. Hover a cell for details.

Affine price curve. The row "heavy-tail · short handling · empirical due dates" has one known search miss: 270 t with two heavy units qualifies with one unit fewer and is cheaper than 550 t in 89 shift-labour decisions.

Sources: this study: 6,480 decisions (paper §5.1, §5.5).

By mass profile: the smallest tier that carries (nearly) every block alone, or light units with a few heavy ones

MASS PROFILE	COVERING TIER	WINNERS (SHARE OF 1,080 DECISIONS)	COUPLES ≤ 1/10	COUPLES NONE	MEDIAN MARGIN
Jiang	500 t	425 t 42.1% · 270 t + 2 heavy 38.1% · 500 t 11.8% · 270 t + 1 heavy 5.6% · 300 t 2.2% · 380 t 0.2%	97.8%	11.8%	1.7%
Light-skewed	500 t	270 t + 2 heavy 59.2% · 500 t 24.2% · 270 t + 1 heavy 16.7%	100%	34.0%	1.6%
Uniform	500 t	500 t 99.3% · 270 t + 2 heavy 0.7%	99.3%	99.3%	2.0%
Heavy tail	—	270 t + 2 heavy 36.6% · 500 t 28.5% · 550 t 22.5% · 380 t 6.4% · 300 t 3.4% · 325 t 1.8% · 425 t 0.8%	47.6%	0%	1.2%
Roh–Cha	550 t	550 t 100%	100%	100%	30.1%
Liu	425 t	425 t 100%	100%	100%	3.3%

- In the five profiles with a covering tier, 99.4% of winners couple ≤ 1/10. In the Jiang sample only 1.3% of blocks exceed 425 t, so 425 t wins
- **The heavy tail is the expected exception** (a menu gap): 9.1% of blocks exceed every tier, so even 550 and 500 t couple 9.1% and 10.8%; of the 145 shift-labour decisions whose winner couples > 1/10, 126 go to 500 t, which couples only the blocks above 500 t
- **Margins are often small**: the runner-up costs a median 2.4% more, under 1% in 18.3% of decisions. Yet outside the heavy tail every fleet within 1% of the least cost couples ≤ 1/10 in 98.8% of decisions (within 2%: 96.8%): the conclusions concern the kind of fleet, not one tonnage
- Quotations: holding every rival and labour cost fixed, overturning the winner takes a capital-price increase of a median 11.1%, and under 5% in 1,123 decisions (17.3%), so purchases need comparable quotations

Sources: this study (paper §5.1, Table 5; Supplementary S9); sources of the mass profiles on “Parameter sources (1)”.

Try it: pick a condition and a price ratio and see which fleet is cheapest

Masses: Uniform | Handling: short handling | Due dates: empirical due dates | Shift labour | Crew-vehicle hours | Crew-team hours

θ 12.4

Fleet	Composition	K*	Daily cost (labour-h)		vs cheapest	Coupled
500 t	5 × 500 t	**5**		**421**	★	0%
550 t	5 × 550 t	**5**		430	+2.0%	0%
270 t + 2 heavy	2 × 500 t + 4 × 270 t	**6**		474	+12.6%	11%
425 t	6 × 425 t	**6**		490	+16.4%	19%
270 t + 1 heavy	1 × 500 t + 6 × 270 t	**7**		543	+28.9%	33%
380 t	7 × 380 t	**7**		561	+33.2%	32%
300 t	8 × 300 t	**8**		619	+47.1%	51%
325 t	8 × 325 t	**8**		626	+48.7%	45%
250 t	9 × 250 t	**9**		681	+61.8%	63%
270 t	9 × 270 t	**9**		688	+63.3%	58%
200 t	12 × 200 t	**12**		888	+110.9%	76%

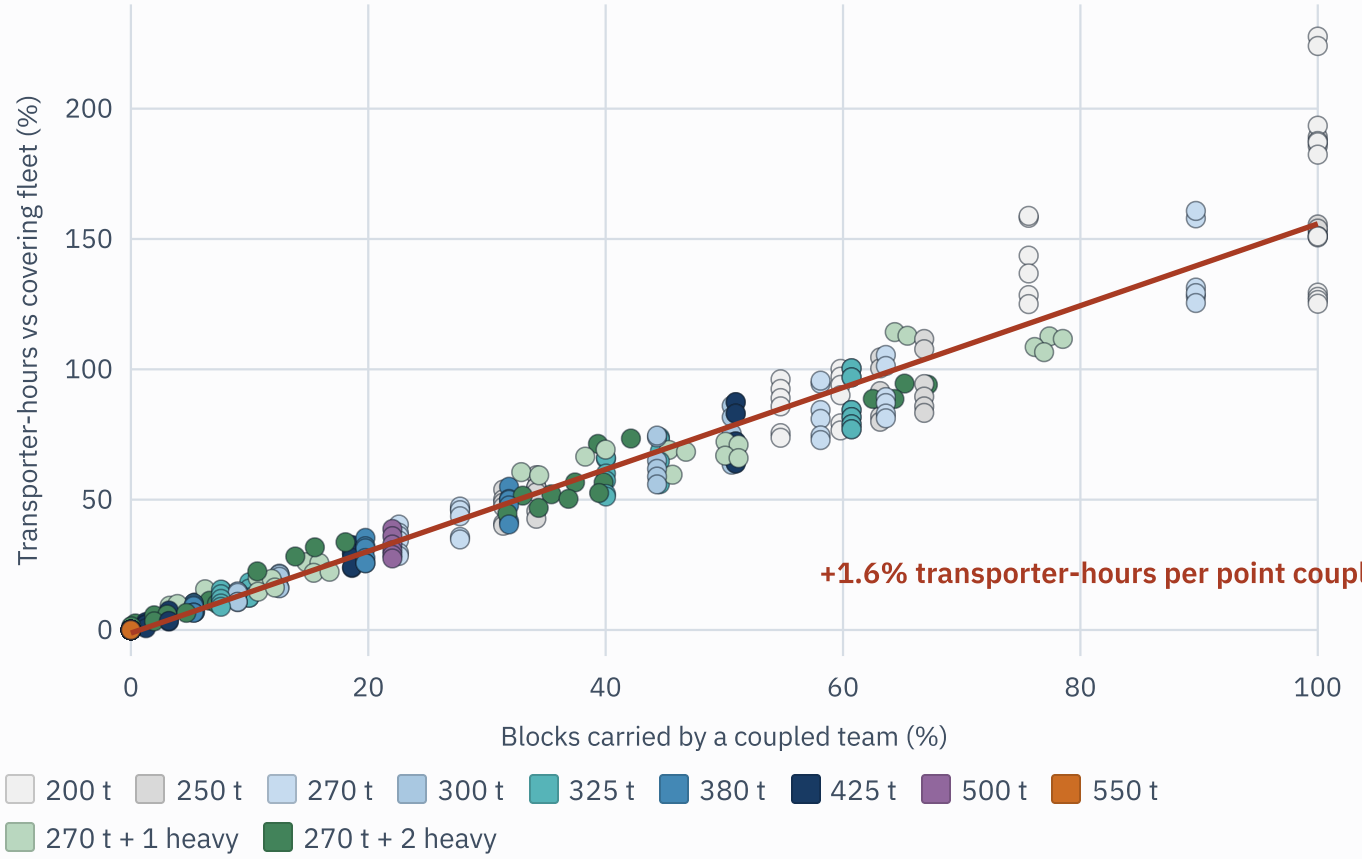
Cheapest: **500 t** (5 vehicles), 421 labour-hours a day; the runner-up, 550 t, costs 2.0% more. Labour is 76% of the cheapest fleet's cost. Capital θ × P | Labour L

Daily cost C = θ × P + L (labour-hours a day). P: fleet price in 270 t equivalents (affine curve); heavy units in mixes are priced at the smallest tier they need. Shift labour = 64 labour-hours per unit; crew measures count only hours actually worked.

Sources: counts and hours: this study; price curve fitted to 33 tender awards; 4 crew × 16 h: Park & Seo (2012), Roh & Cha (2011) (paper §3.2).

Question (ii): excess coupled-service time, repeated service plus alignment, makes up most of the extra transporter-time; partner waiting is small

330 sized fleets outside the heavy-tail profile: transporter-hours a day above the covering fleet of the same condition



+1.6%

transporter-hours a day for each percentage point of blocks coupled (r = 0.97)

2.2×

the covering fleet's transporter-hours, for fleets that couple 60–100% of blocks

0.55%

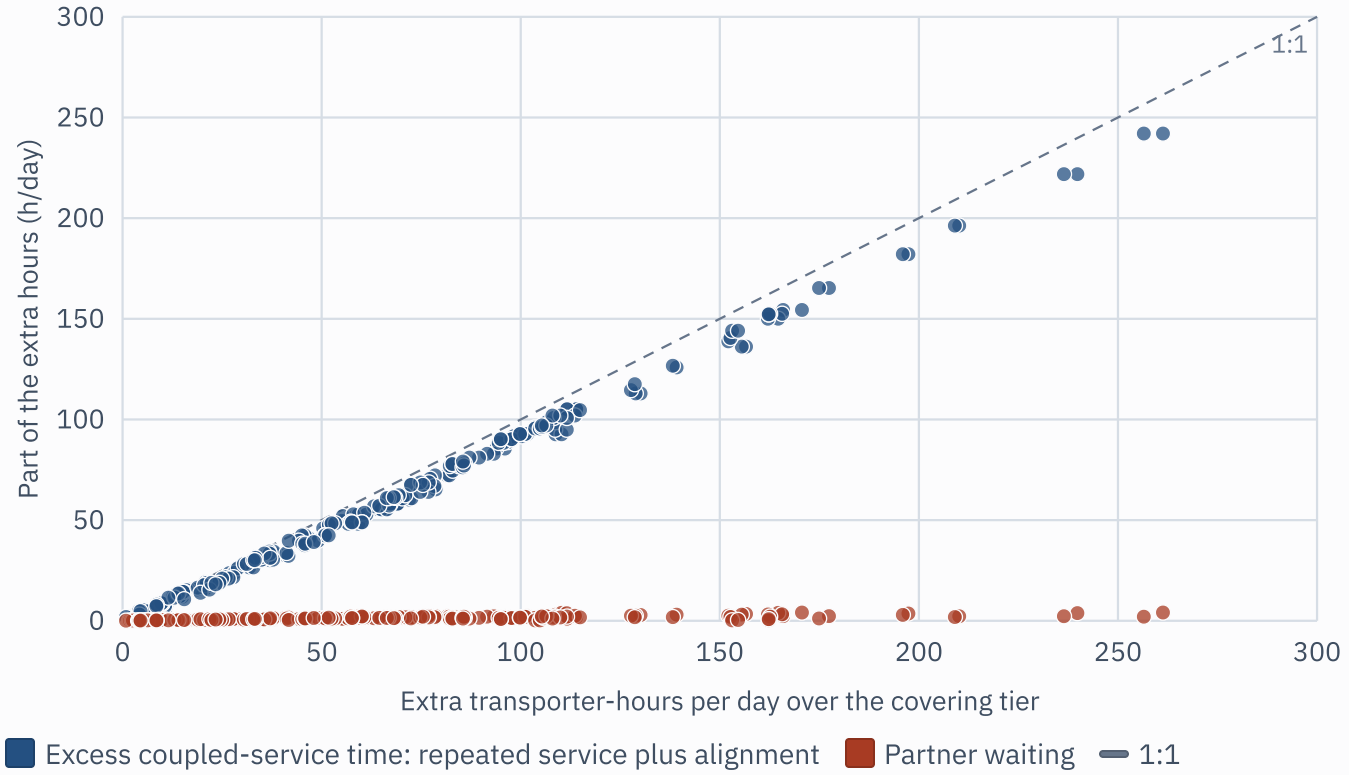
waiting for the last team member: a median 0.55% of working time, never more than 2.2% for any fleet

Partner waiting is small; the cost is that two or three vehicles and crews are held for the whole loaded trip, alignment included, and leave parallel service.

Sources: this study: working and waiting hours of each sized fleet (paper §5.2, Fig. 8a).

Block by block: a median 90% of a light fleet's extra transporter-time is excess coupled-service time (repeated service plus alignment), 2% partner waiting

266 lighter fleets against the covering tier of their condition (outside the heavy tail, coupling > 1% of blocks): extra transporter-hours and their two parts



89.8%

median share of the extra transporter-time that is excess coupled-service time, (|S| − 1)·D + |S|·δ per block: repeated service plus alignment (10th–90th percentile 77.5–93.8%)

2.1%

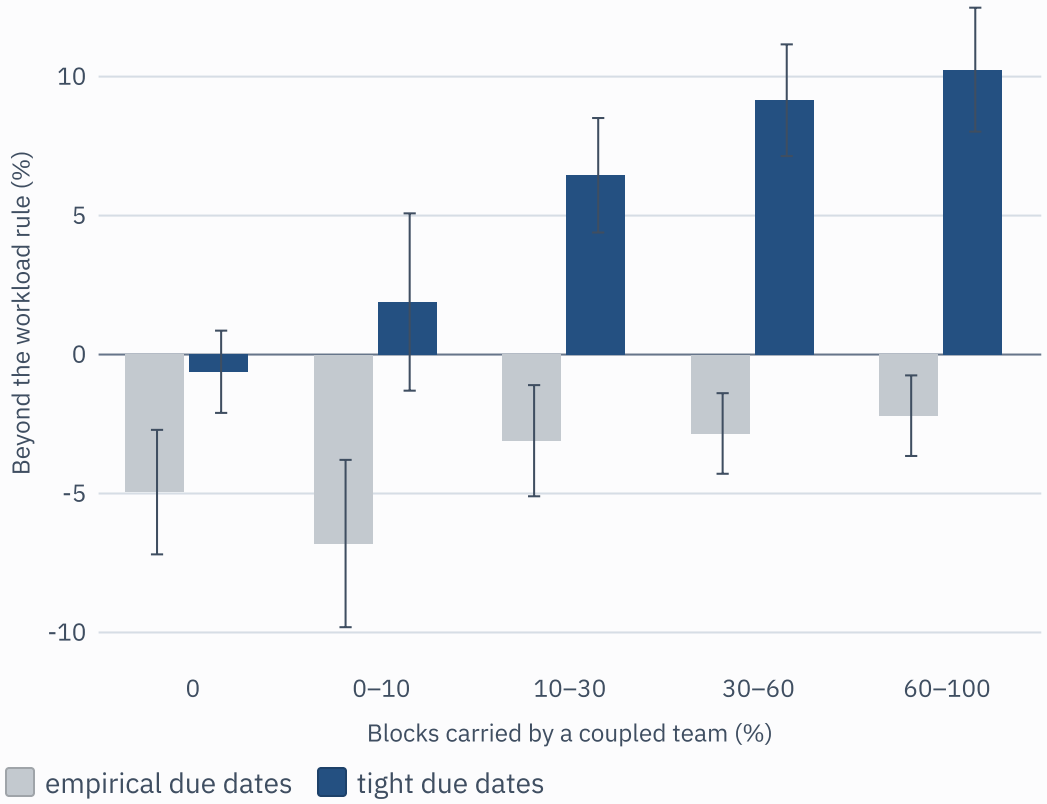
median share spent waiting for partners (1.3–3.2%); early members could leave later, so this is an upper bound

- The rest is the difference in empty travel, which can be negative: points above the 1:1 line have less empty travel than the covering tier
- The same decomposition gives the covering fleet's occupancy per block used by the budget: about 43 min short, 63–85 mass-dependent, about 80 long loading

Sources: this study: block-by-block decomposition of the priced schedules, lighter fleets of the 384 sized fleets outside the heavy tail that couple more than 1% of blocks (paper §5.2, Fig. 8b).

The fluid screen rules out the lighter tier in 77% of comparisons with one error; scheduling settles the rest

Transporters needed beyond the workload rule K_ρ (share of K*, mean with 95% interval)



r = 0.97

fluid occupancy factor against the simulated count ratio K^*_{lo} / K^*_{hi} (210 fleet pairs)

77% · 1

with a 10% margin (fluid cost ratio > 1.10) the screen rules out the lighter tier in 7,311 of 9,450 comparisons; the simulation finds it cheaper in only one, by 0.1%

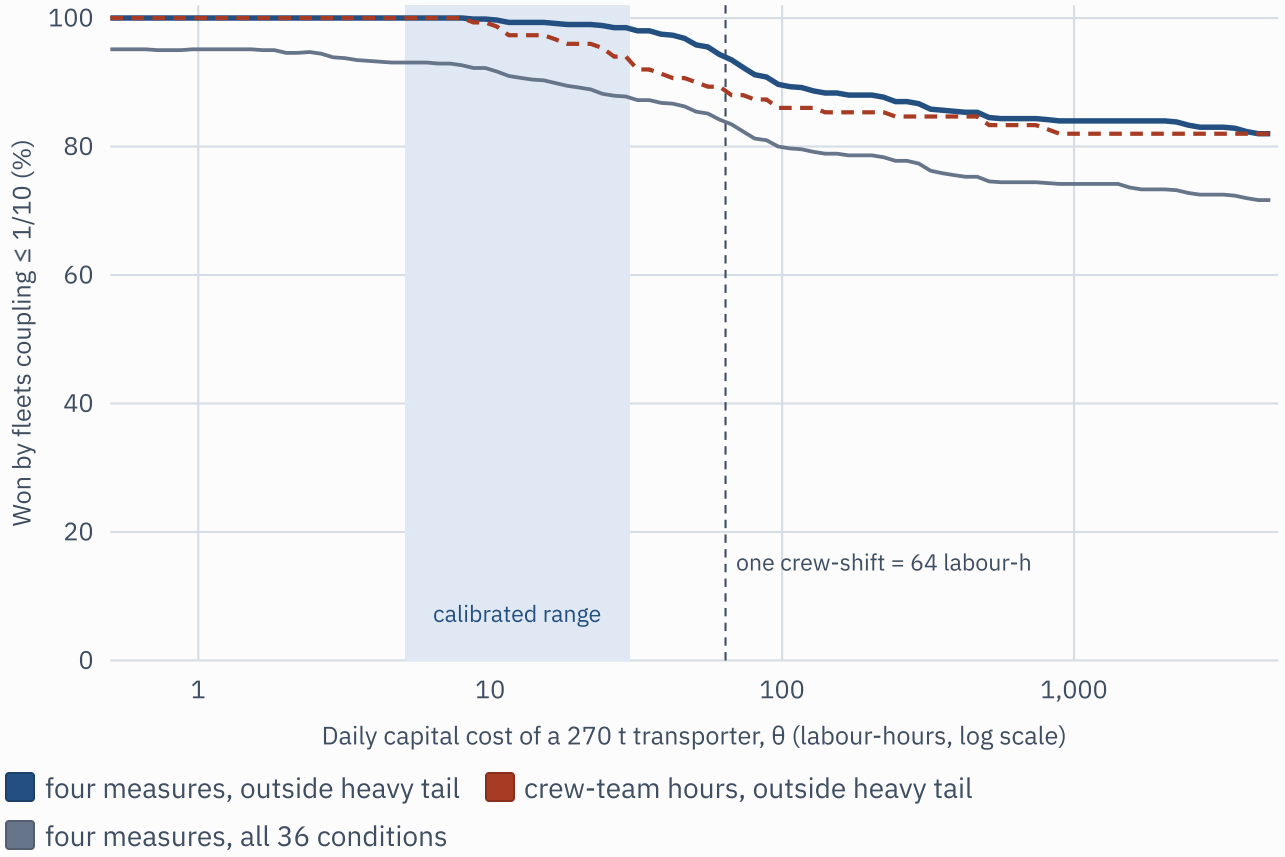
- Of the 571 comparisons the lighter tier wins, 570 lie among the 2,139 left to scheduling; a 5% margin rules out 82.0% with 35 errors, no margin 92.3% with 274
- K_ρ is within one of K* for 319 of 384 fleets (all 192 with empirical due dates); under tight due dates fleets coupling > 60% need 10.2% of K* more, those coupling none 0.6% fewer: a coupled lift needs several vehicles free at once
- Priced on K_ρ: low-coupling share 92.0% (against 91.0%) but the same winner in only 57.7%; 17.7% of the selections lack a qualifying schedule

> **The 77% is a within-sample check: occupancy comes from covering-fleet schedules, and a screen from inputs alone is untested.**

Sources: this study: every lighter homogeneous tier against the covering tier in each condition that has one, × 5 price curves × 9 values of θ (shift labour; paper §5.2, Fig. 8c; Supplementary S9).

Coupling-reliant fleets could win only if capital became far dearer than crews

Share of decisions won by fleets coupling ≤ 1/10 as θ runs from 0.5 to 5,000 (conditions × 5 price curves × the labour measures named)



With counts and schedules fixed, every fleet's cost is a straight line in θ, so the winner at any θ follows exactly from the lower envelope.

100% · 98.3%

outside the heavy tail: every decision up to θ ≈ 8.7, and 98.3% at θ = 30

82%

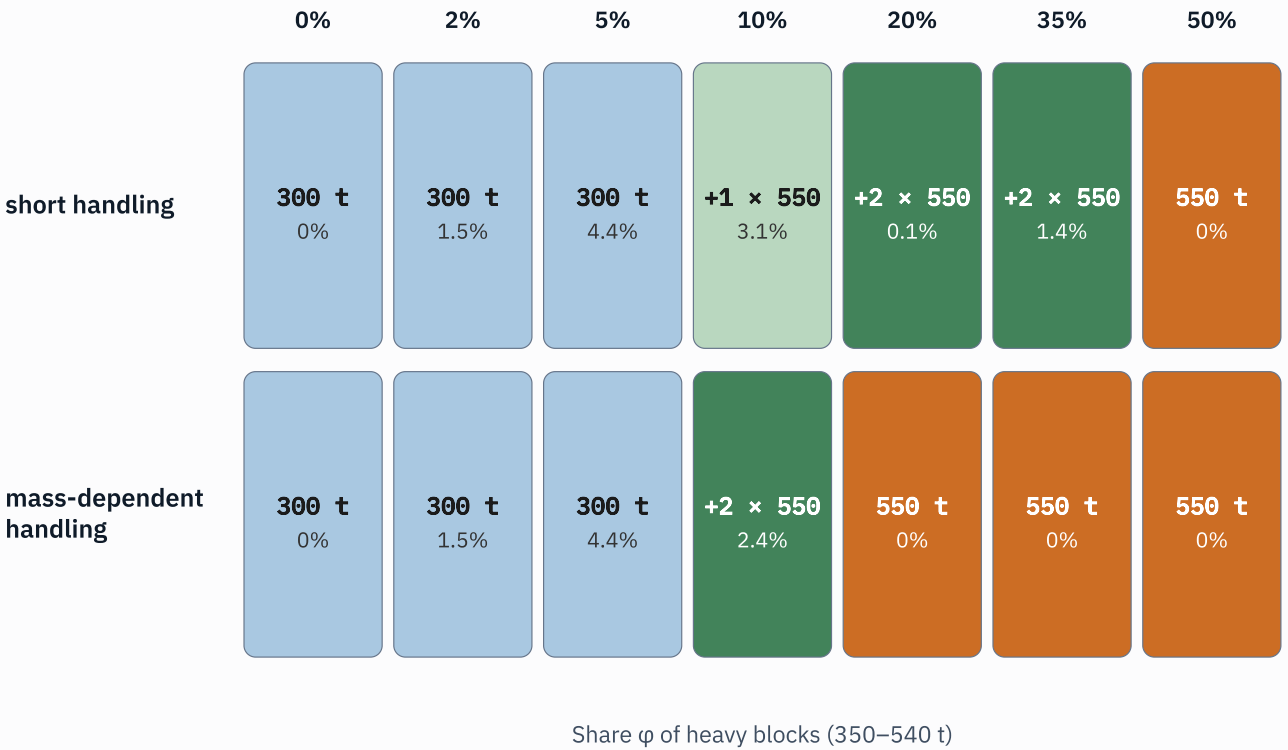
even with capital alone (θ → ∞): sub-additive prices make a single carrier cheaper per block without crews (Proposition 1)

- In 82% of the 600 combinations no θ up to 5,000 makes a coupling-reliant fleet cheapest; 98 of the other 108 switch above θ = 30 (median 87 labour-h, ~1.4 crew-days)
- The few switches inside the calibrated range all come under crew-team hours (dashed), which assume one crew drives a coupled team
- A crew-day now costs 2.1–12.5 times a 270 t unit's daily capital; driverless transporters could approach the switch

Sources: this study (paper §5.2, Fig. 8d); the calibrated range 5.1–30 comes from tender prices, official wages and the assumed capital-recovery settings (the two parameter slides).

Question (i): as heavy blocks become common, the cheapest fleet moves from light to light-plus-heavy to heavy, always coupling little

Cheapest fleet by the share φ of heavy blocks (350–540 t; the rest 100–300 t), affine price · shift labour; the same at all nine price ratios



- 300 t wins every decision at φ = 0 and 92–97% at φ = 2%; from φ = 5–10% light units with one or two heavy units take over
- 550 t first wins at φ = 50% with short and at φ = 20% with mass-dependent handling (all 45 shift-labour decisions); the true switch lies between these points and the next lower ones
- The budget explains both switches: 300 t stays within its budget up to φ ≈ 5%; the mixes lose to 550 t once the heavy workload exceeds two heavy units
- The simple formula ranks 300 vs 550 t correctly in 94.4% of 630 comparisons

2,502 / 2,520

decisions where the cheapest fleet couples at most one block in ten; the exceptions are the two-heavy mix at φ = 35% with mass-dependent handling under per-team crews (19.6%)

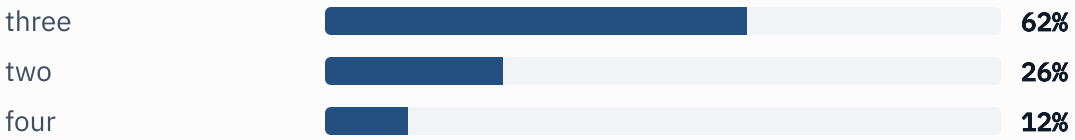
The winners' mean coupled share is at most 3.3% at every φ.

Sources: this study: 30 further days with φ from 0 to 50% (paper §5.3, Fig. 9); heavy 350–540 t and the rest 100–300 t are assumed ranges.

Richer mixes change the cheapest fleet but not the principle: 300 t plus 425 t units win all 8,000 cost scenarios for the Liu sample

Liu sample, short handling, 120-min cap; candidates = the 11 reference fleets plus extended mixes; 5 price curves × 400 values of θ × 4 labour measures = 8,000 cost scenarios.

Heavy 425 t units in the cheapest mix (share of scenarios)



44% of the Liu blocks exceed 300 t, so a mass-matched five-vehicle fleet would hold about 2.2 heavy units; the cheapest mixes couple at most 2.1% of blocks.

0.3–7.7% is how much more the best other candidate costs

1.3–7.7%

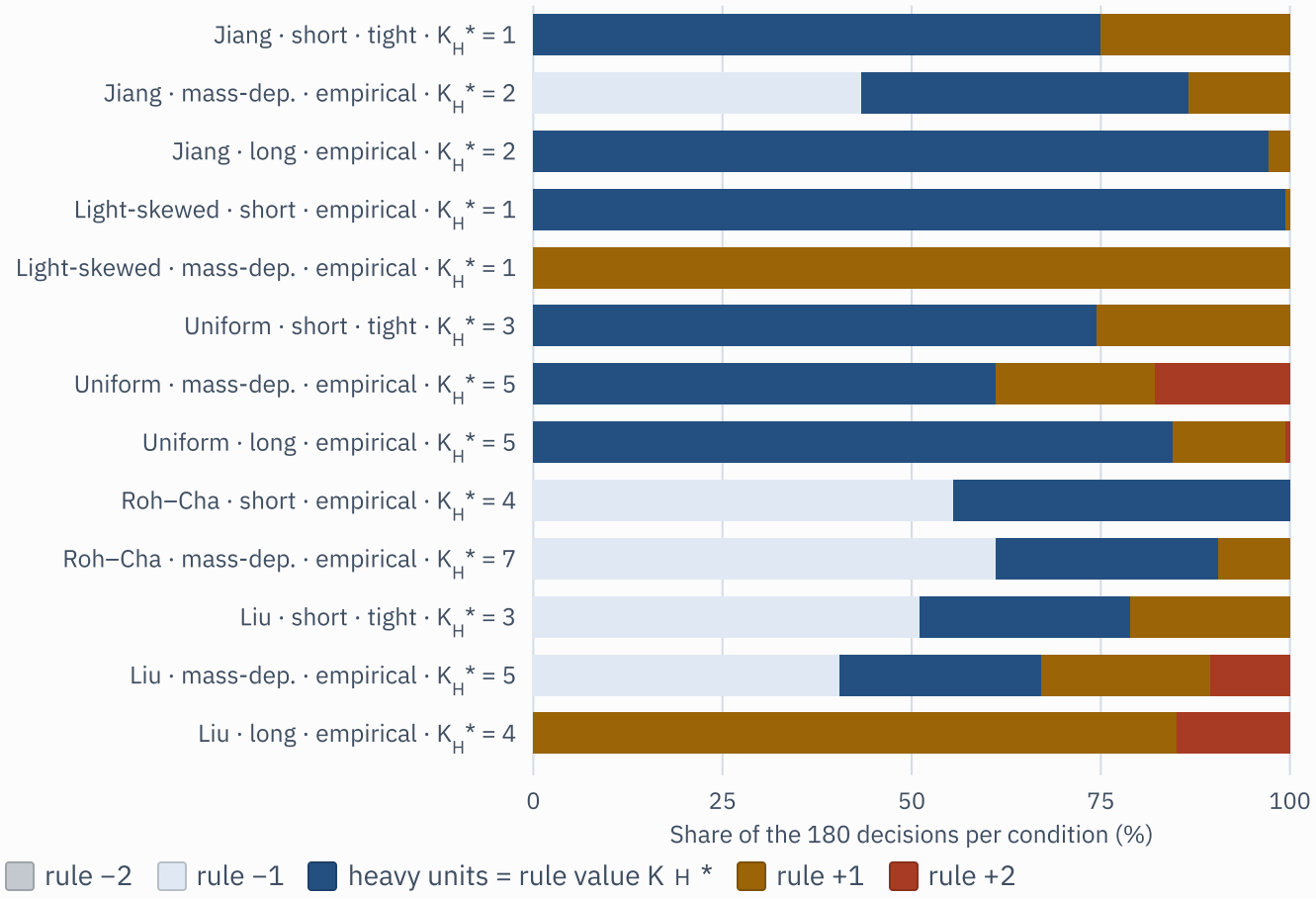
is how much more the homogeneous 425 t fleet, the winner among the reference types, now costs under shift labour

> **The number of heavy units follows the heavy workload; richer mixes change which fleet is cheapest, not the single-carry principle.**

Sources: this study: the extended candidates of the lateness-cap study (paper §5.3); Liu masses from Liu et al. (2022), S1 data.

In 13 further conditions, light units with matched heavy units win 95.3% of decisions

Heavy units of the cheapest 300 t mix vs. the rule value $K_H^* = \lceil K_{cov} \cdot h_H \rceil$



95.3%

of 2,340 decisions go to 300 t units with covering heavy units, always coupling ≤ 1/10; the best reference fleet costs a median 1.9% more (at most 10.2%). The light-skewed exceptions go to 270 t units with one or two heavy units, also light–heavy mixes

96.6%

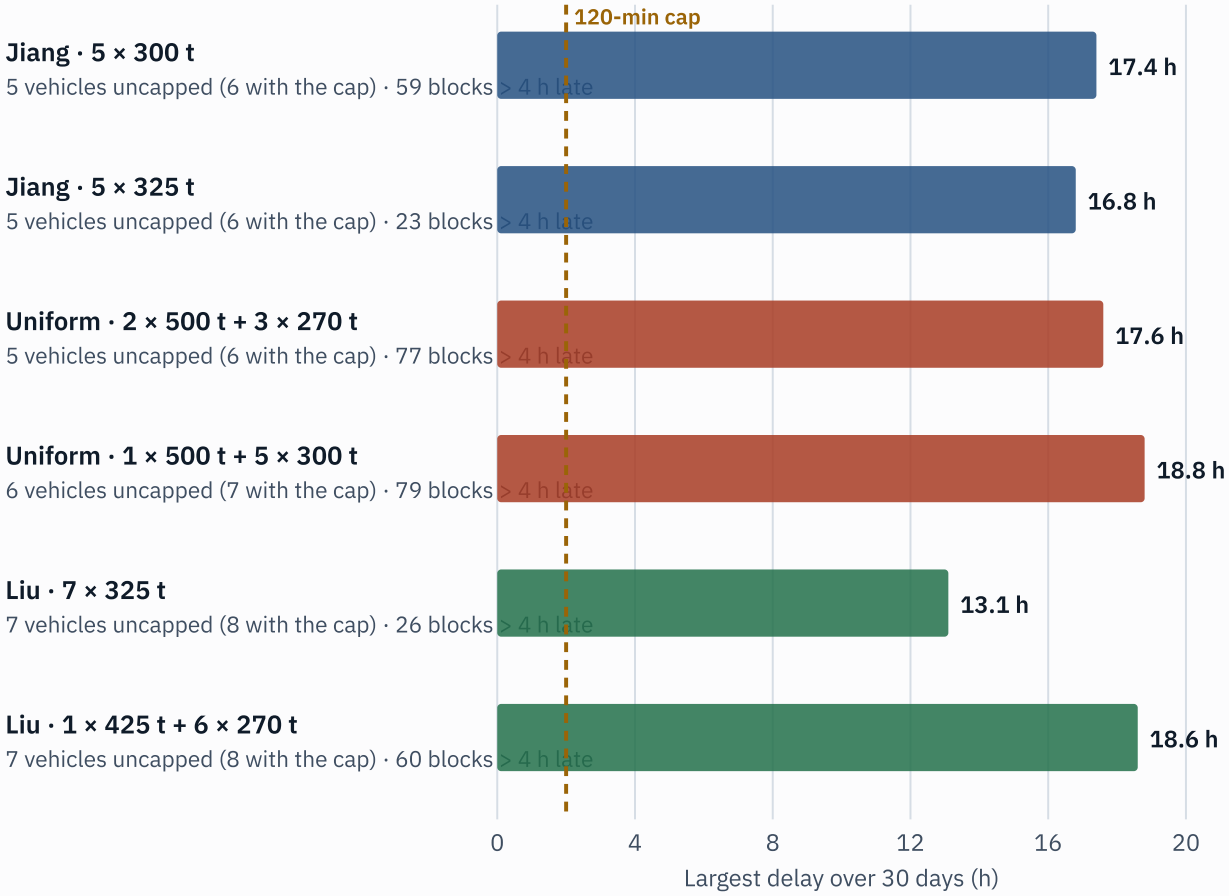
have a heavy count within one unit of K_H^*; the most frequent count equals K_H^* in 6 of the 13 conditions and is one off in the others

- Too few heavy units show the budget at work: uniform profile, tight due dates, one 500 t unit leaves 27% of blocks coupled and needs seven transporters, while three need five and couple 0.3%
- In four conditions 79 decisions (3.4%) pick $K_H^* + 2$, the largest tested; in three the next count up is the covering fleet, already a candidate, so only the Liu sample with long loading (1.2%) could hide a cheaper count

Sources: this study: 13 further conditions, 300 t units with $K_H^* - 2$ to $K_H^* + 2$ covering heavy units (500, 550 or 425 t) (paper §5.3; Supplementary S5).

Question (iii): with an on-time share alone, fleets qualify with one unit fewer by deferring their hardest blocks 13–19 h, so service needs a lateness cap

Fleets that need one unit fewer without a cap: the latest block of each over 30 days



6

sized fleets (five fleet types in three conditions) qualify with one unit fewer without a cap: their latest block is 13–19 h late

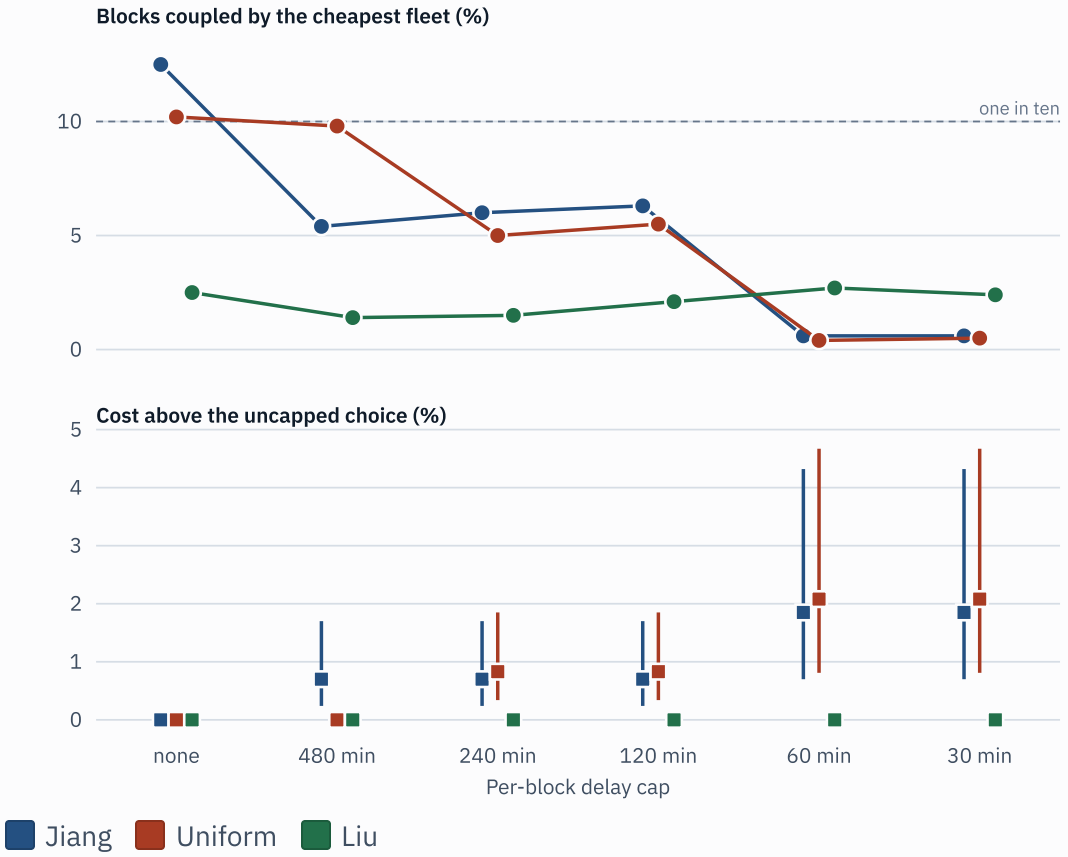
The deferred blocks are mostly coupled ones: of the blocks more than 4 h late, 40 of 59 (300 t) and 16 of 23 (325 t) under Jiang masses, and 24 of 26 (325 t) under Liu masses. In the mixes they are blocks queued for the scarce heavy units.

With an average target alone, coupled lifts can be pushed back indefinitely, which no production plan can absorb; tighter caps never lower a count.

Sources: this study: caps from 30 min to none, three mass profiles with short handling (paper §5.4, Fig. 10a).

With a 120-min cap, fleets that qualify by deferring are excluded, and least cost rises by only 0–1.8%

Shift labour; short handling, empirical due dates; 30 days per mass profile



Masses	No cap	120-min cap	Cost
Jiang	5 × 300 t 12.5%	1 × 500 t + 4 × 270 t 6.3%	+0.2–1.7%
Uniform	2 × 500 t + 3 × 270 t 10.2%	2 × 500 t + 3 × 300 t 5.5%	+0.3–1.9%
Liu	2 × 425 t + 3 × 300 t 2.5%	2 × 425 t + 3 × 300 t 2.1%	0

- Under Jiang and uniform masses the cap changes the purchase: to mixes with heavy units that couple less
- Without a cap the winner couples ≤ 1/10 in only 75% of their decisions; with any cap from 480 down to 30 min, in all of them
- Tightened to 30 min, cost rises by at most 4.7%; under Liu masses 2 × 425 t + 3 × 300 t is cheapest at every cap

Sources: this study (paper §5.4, Fig. 10b, c).

The on-time target matters far less than the cap: at 90% or 98% the count barely moves

40 · 37

of 42 fleets keep their K* at a 90% · 98% target; the others move by one transporter

93.3% • 87.1%

of 1,260 decisions keep the 95% winner; outside the heavy tail every winner couples ≤ 1/10 at both levels

Why: the integer step of one transporter

- At K* the best schedules already keep at least 98% of blocks on time for 36 of the 42 fleets (a 37th falls one block short)
- One transporter fewer drops the pooled on-time share to 56–96%, and leaves at least one day without a schedule within the cap for 32 of the 41 fleets that can lose one

An average target cannot replace a limit on extreme lateness.

Sources: this study: 42 fleets in seven conditions, re-sized at 90% and 98% on the original 30 days (paper §5.4).

Flexible coupling adds little at the fleet level; only zero alignment time lets a light fleet win

Two coupling rules

Flexible (main rule): any minimal team. **Restricted**: mixes may couple only blocks that no transporter of the fleet can carry alone. Counts found under the restricted rule are pooled into the flexible results, since its schedules remain feasible.



Change of least cost under the restricted rule (shift labour, 1,620 decisions); at most 9.4%. Zero for the uniform, heavy-tail, Roh–Cha and Liu profiles, largest for light-skewed (mean 1.4%). The mixes need flexible coupling (only 6 of 72 still qualify), but the decision hardly changes, because a single-carry fleet is nearly as cheap. Without pooling, search noise would make the restricted rule look cheaper in 8.6% of comparisons.

Alignment time

Jiang sample, short handling, empirical due dates, six fleet types: with no alignment time (δ = 0) a 300 t fleet coupling 12.5% wins 77 of 180 cost scenarios; from δ = 2.5 min it wins none, and the winner becomes 270 t with one heavy unit (coupling 6.3–6.9%), then with two heavy units from δ = 20 min (≤ 0.2%).

88.6% → 100%

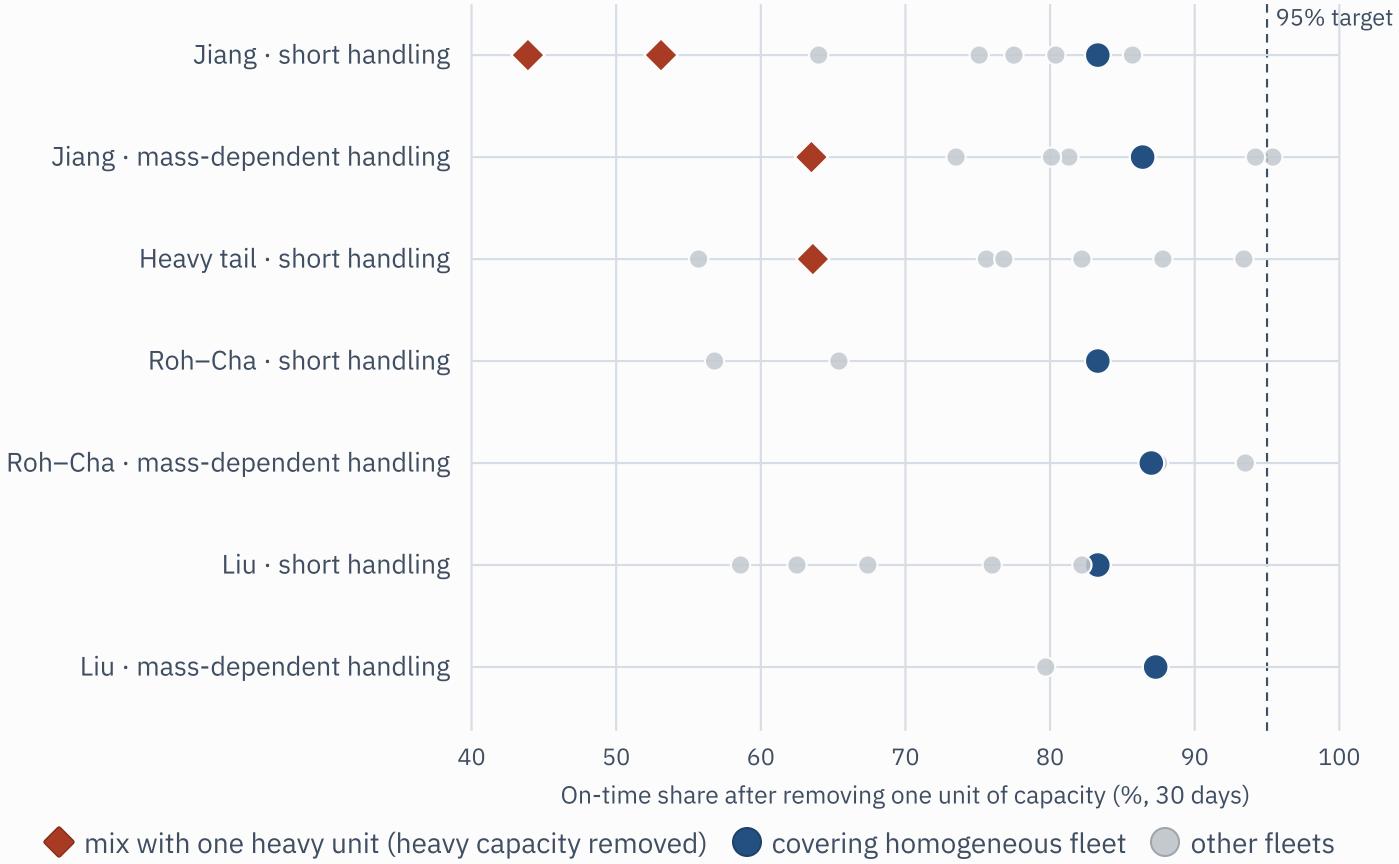
low-coupling winners in the one-factor runs (reduced candidate set): 88.6% at δ = 0, 100% at δ = 20–40 min

> **Coupling pays only when it is almost free.**

Sources: this study (paper §5.4; Supplementary S2); the restricted rule as in Jiang et al. (2021); alignment time has no published measurement, 0–40 min is the tested range.

Capacity-removal estimate: least-cost redundancy is a second heavy unit or one more transporter, not coupling

On-time share over the 30 days after removing one unit of capacity (one fewer in a capacity class, start positions reassigned; 42 contending fleets in seven conditions)



44–64% · 83–87%

outside the heavy tail: single-heavy mixes without their heavy capacity class · covering homogeneous fleets with one fewer unit

- Redundancy is estimated from the qualifying counts of the original and reduced compositions. At those counts, normal-operation schedules give 87.5% of 6,480 winners coupling ≤ 1/10 (98.8% outside the heavy tail), at a median 8.8% above the nominal winner; outside the heavy tail no single-heavy mix wins (5.8% of nominal winners)
- The option to couple is worth about as much as nominally (90th-percentile upper bounds 6.4–6.7%); fleets with more units lost relatively less service in only 3 of 7 conditions

This is a capacity-composition estimate: start positions are reassigned rather than kept for the surviving vehicles, so it does not establish service after an arbitrary vehicle fails.

Sources: this study: reduced compositions rescheduled on the 30 original days with the start positions of their own scans; redundancy counts estimated from the stored runs (paper §5.4; Supplementary S2, S5).

How reliable are the results? We looked for what could overturn them

① Other working days

Draw 30 days from the 30 **with replacement** and compare all fleets again; repeat 2,000 times (the bootstrap) and count how often the result holds.

1	2	3	4	5	6	7	8	9	10	11	12	13	14
15	16	17	18	19	20	21	22	23	24	25	26	27	28
29	30												

1	1	2	3	3	3	5	5	5	7	7	8	9	10
10	11	18	18	18	19	19	21	22	23	24	24	26	28
29	30												

Top: the original 30 days; bottom: one redraw, yellow = days drawn more than once

② Distance to a flip

Counts come from search: with more search a loser might manage with one unit fewer and overtake. We compute how many on-time blocks it still lacks at one fewer and compare that with what two extra searches recover.

strong: short by > 92 **medium: 59–92**

weak: ≤ 58

92 = the most on-time blocks two extra searches recovered in any fleets (30 days); 58.9 = its 95th percentile

③ Disturb the inputs

Multiply every loading and unloading time by a random 0.5–1.5 (±50%) in 20 scenarios a day, to see whether fleets sized to the target still suffice, then re-size them to meet it under noise.

$$\text{handling} \times U(0.5, 1.5) \cdot 20 + 20 \text{ scenarios a day}$$

Also checked: exact counts on small instances, 30 fresh days, six more searches a day at the boundary, counts from shallow search, the worst case of every coupling fleet losing a unit, unmeasured parameters, manufacturer speeds, capacity removal and a second yard. All 214,210 scheduling runs are kept and can be recomputed.

Sources: bootstrap Efron & Tibshirani (1993); grading thresholds and noise tests are ours (paper §3.4; Supplementary S2, S4); ±50% is an assumed noise level.

Checks of the findings: they change which fleet is cheapest, not whether it carries alone

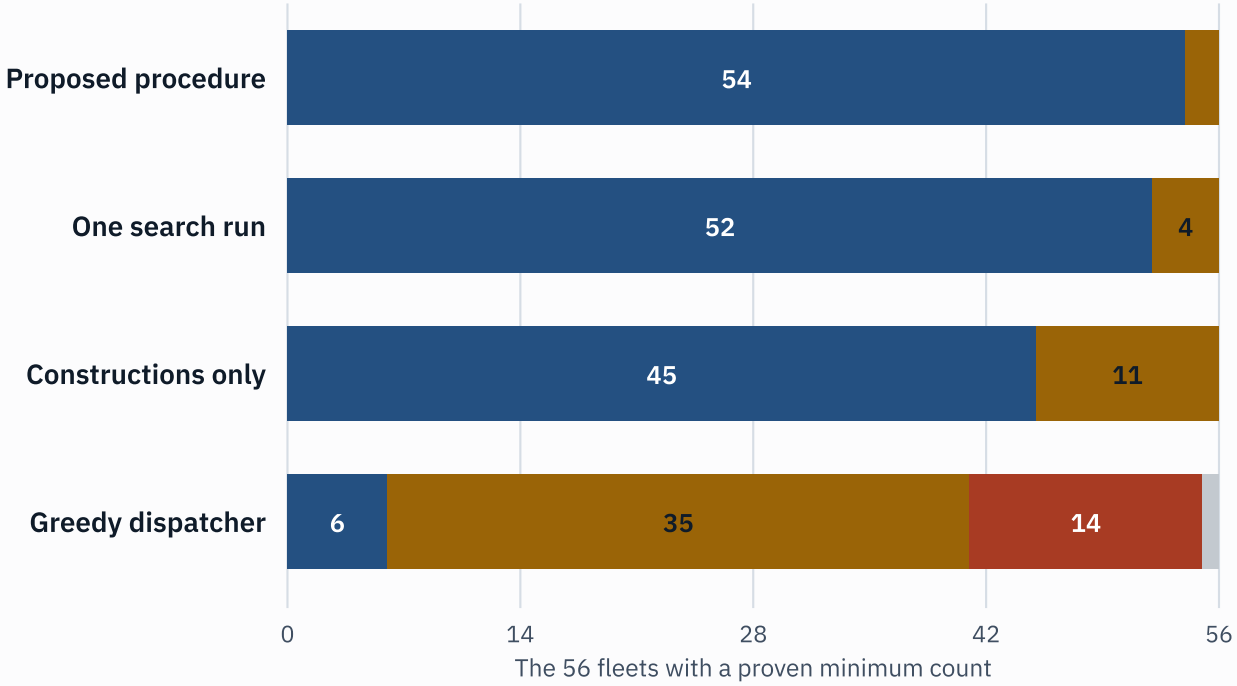
CHECK	WHAT CHANGES	LOW-COUPLING SHARE (OUTSIDE THE HEAVY TAIL)	OTHER OUTCOME
As reported, shift labour	—	91.0% (100%)	1,620 decisions
As reported, all labour measures	four labour measures	90.8% (99.4%)	6,480 decisions
Search depth	counts after 0, 100 and 300 iterations; one search a day	90.2 · 88.8 · 89.3 · 90.8%	same winner 77.5 · 81.1 · 89.4 · 95.0%
Boundary re-search	six more searches a day at K* − 1 for 66 fleets and 10 controls	90.8%	K* fell for 3 fleets (no control); 242 decisions (3.7%) changed
Exact benchmark	80 single-batch slices; 70 fleets in 8 conditions	—	count equal to the exact count for 54 of 56 proven fleets, one higher for 2; certified winner in all 360 shift-labour decisions
Adversarial bound	all 258 coupling-reliant fleets given one transporter fewer	68.8% (80.9%)	worst case
Day resampling	2,000 bootstrap samples of the 30 days, stored schedules re-used	89.2% (> 99.9%)	K* unchanged in 89.0% of resamples on average (median 97%)
Fresh days	30 new days; 42 fleets in 7 conditions	98.6% (98.3%)	same winner in 86.5% of 1,260 decisions; K* within one for all 42
Slow speed and long handling	425 t, 500 t, mixes; handling ×1.5, ×2	100% of 3,060 resolved	540 decisions outside the specification for every candidate
Capacity removal	reduced compositions, starts reassigned	87.5% (98.8%)	same winner in 78.6%; least cost +8.8% (median)
Second case	Liu et al. (2022) data, sized from scratch	100%	425 t cheapest in 179 of 180 decisions

Search dependence is deliberately strict: a decision counts if any loser would be cheaper with one transporter fewer (5,033 of 6,480, 77.7%); this is not a share of wrong decisions. Graded at K* − 1: 3,513 strong, 298 medium, 1,222 weak. All 1,447 certain decisions have a winner coupling ≤ 1/10, against 88.1% of the search-dependent ones.

Sources: this study (paper §5.5, Table 6; Supplementary Table S1).

Against exact counts on small instances: the procedure finds the proven minimum for 54 of 56 fleets and is never below it

Each method's count against the proven minimum (single-batch slices, the 56 fleets with a proven minimum)



54 / 56

fleets whose count equals the proven minimum; 2 are one higher, none lower

360 / 360

shift-labour decisions pick a certified winner, although all are close

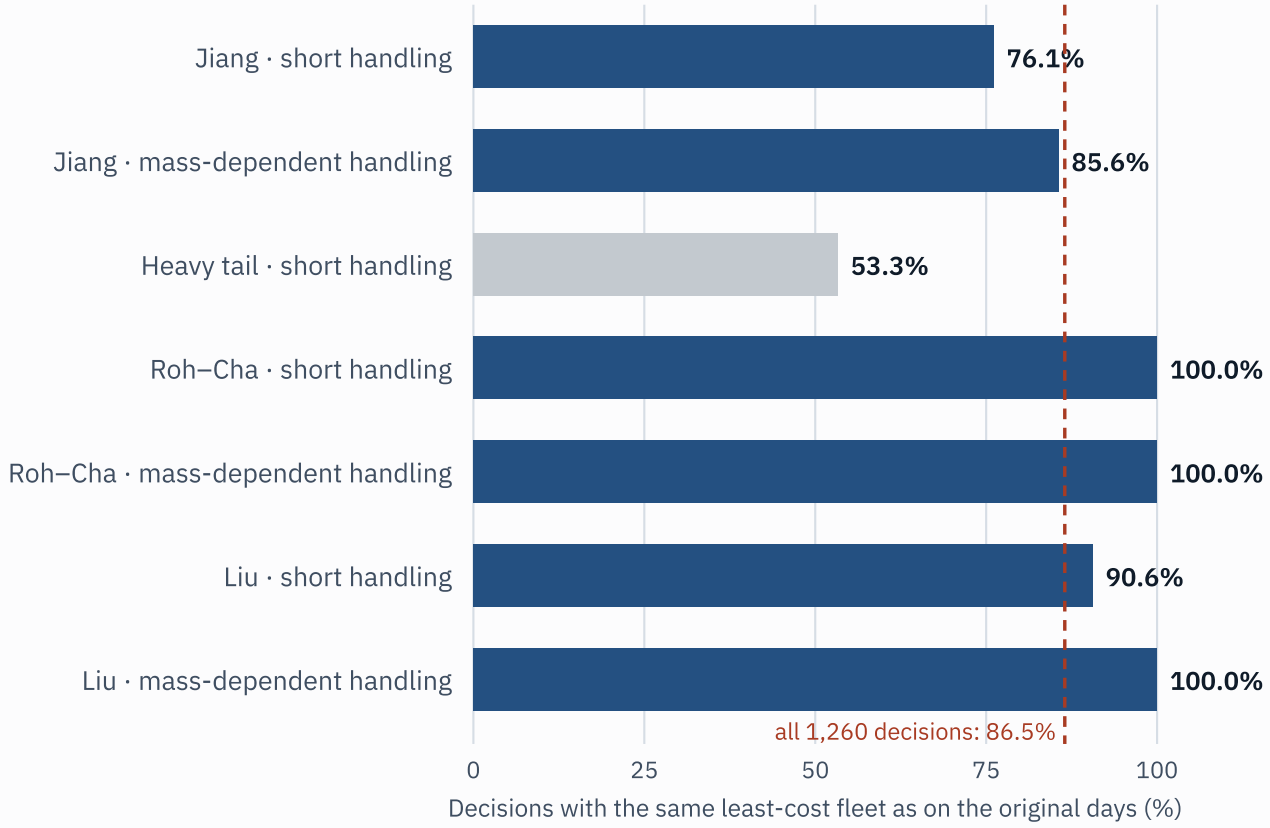
- One search run finds the fewest late blocks on 94.1% of 1,147 proven slice–count pairs, never more than one above
- Constructions alone or a greedy dispatcher: counts too high for 11 and 49 of 56 fleets (greedy: one more without a count); cheapest fleet changed in 37.5% and 25% of decisions, at up to 4.5% and 11.9% more cost

Good scheduling matters for fleet design: cruder dispatching overstates the fleet.

Sources: this study: 80 single-batch slices of 12–13 blocks from ten days in eight conditions; CP-SAT of Google OR-Tools computes the fewest late blocks at each count (paper §4.2, §5.5; Supplementary S7).

On 30 fresh days the kind of fleet holds; the named winner is the same in 86.5% of decisions

Share of each condition's 180 decisions whose least-cost fleet is the same as on the original days



39 / 42

fleets keep their K*; the rest stay within one transporter (two fell, one rose)

98.3%

of the 1,080 decisions outside the heavy tail have a winner coupling ≤ 1/10 (98.6% of all 1,260)

- The largest change is in the heavy tail, where a 500 t fleet needed one transporter fewer and wins 84 of the 180 cost scenarios that 550 t or 270 t with two heavy units had won
- Holding the original counts, 41 of 42 fleets still have qualifying schedules on the fresh days; the exception is Roh–Cha 500 t with short handling: six transporters have no capped schedule on one day, seven qualify

Sources: this study: 30 days not used anywhere else; the 42 fleets contending for least cost in seven conditions, same service specification (paper §5.5; Supplementary S5).

Manufacturer speeds by model: 97% still couple ≤ 1/10; only under Jiang masses with short handling does another heavy-unit mix win

Qualifying count K*: common speed → manufacturer speeds (yellow = changed; grey = not rerun, count kept)

Condition (empirical due dates)	380 t	425 t	500 t	550 t	270 t + 1 heavy	270 t + 2 heavy
Jiang · short	**5 → 6**	5 → 5	5 → 5	5 → 5	**5 → 6**	5 → 5
Jiang · mass-dep.	8 → 8	7 → 7	7 → 7	7 → 7	9 → 9	8 → 8
Roh–Cha · short	**8 → 9**	8*	6*	5 → 5	10*	**8 → 9**
Roh–Cha · mass-dep.	16 → 16	16*	12*	9 → 9	19*	**17 → 18**
Liu · short	**6 → 7**	5 → 5	5 → 5	5 → 5	**8 → 9**	**6 → 7**
Liu · mass-dep.	**10 → 11**	8 → 8	8 → 8	8 → 8	14*	**12 → 13**

380 and 425 t at 10/5 km/h, 500 t, 550 t and the mixes at 12/5 km/h. A faster vehicle never makes a block later, so common speeds are the optimistic end and mixes at the heavy speed the pessimistic end.

45 / 270

shift-labour decisions change their cheapest fleet, all with Jiang masses and short handling: the one-heavy mix needs one more vehicle and the two-heavy mix replaces it; keeping the old choice costs up to 18.6% more

178 / 1,080

decisions change their cheapest fleet at the pessimistic end (114 at the optimistic end)

1,054 / 1,080

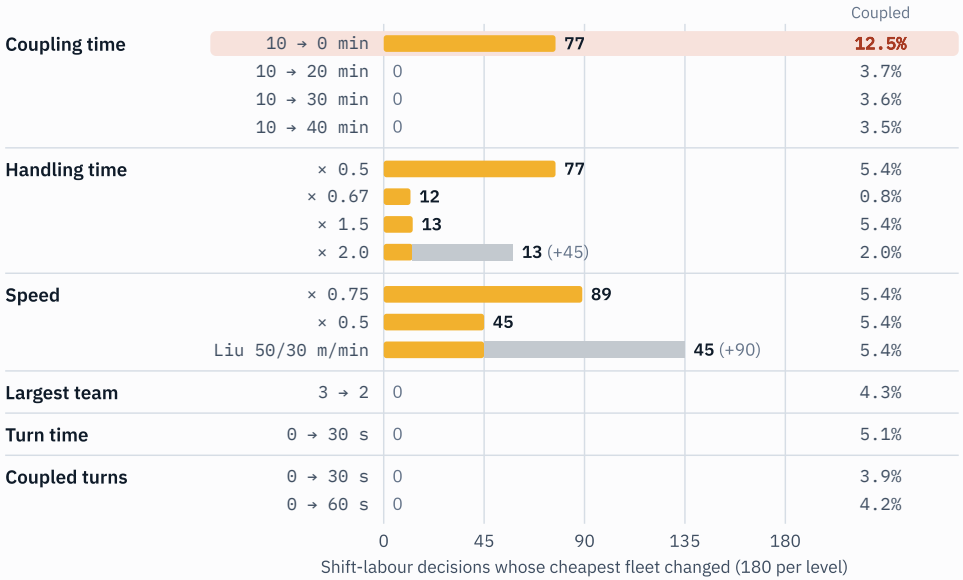
decisions where the cheapest fleet still couples at most one block in ten (pessimistic end)

- Under Roh–Cha masses 550 t is cheapest in all 360 decisions; under Liu masses 425 t in 350 of 360 (among the eleven standard fleets)

Sources: speeds Suzhou Dafang data sheets DCY270 (12/6), DCY380 (10/5), DCY500 (12/5 km/h); 425 t and 550 t take the nearest model's speeds (assumed). Results: this study (paper §5.6; Supplementary Table S1).

Unmeasured parameters one at a time: the choice moves only between neighbouring fleets; only zero alignment time lifts coupling above one in ten

Settings whose cheapest fleet changed when one parameter changed (affine price · shift labour; right: the largest share the cheapest fleet coupled)



Grey: no fleet qualified within four units above its starting count (all mass-dependent handling). With 425 t, 500 t and mixes added and scans up to 40 transporters, every resolved winner couples at most 6.2% of blocks (3,060 decisions); in the other 540 no candidate meets the specification at any count up to 40, so they lie outside the conclusions.

98.3%

of 10,260 decided decisions: the cheapest fleet couples ≤ 1/10

12.5%

zero alignment time: 300 t becomes cheapest (Jiang masses, short handling), coupling 12.5%; from 2.5 min a mix with heavy units wins

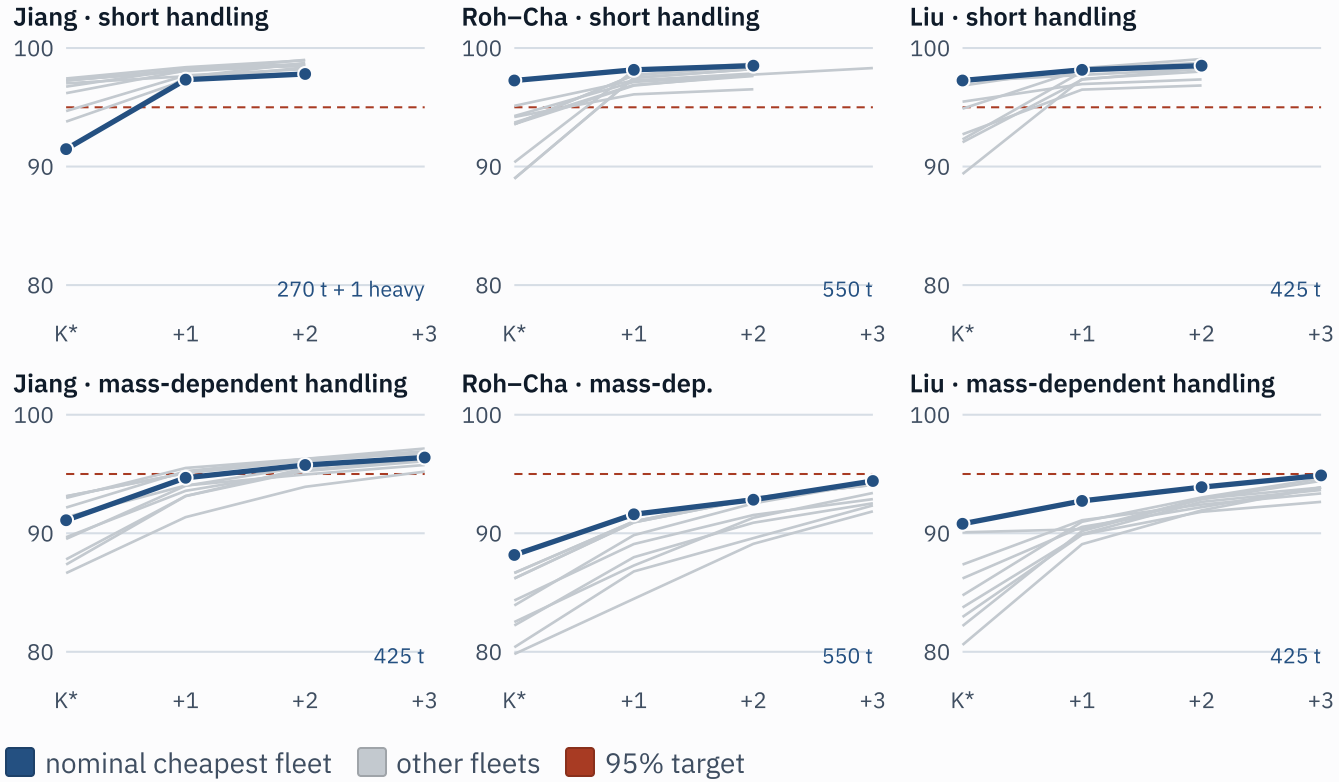
- Every change is under Jiang masses, between neighbouring fleets; under Roh–Cha masses 550 t never changes
- With one crew per coupled team, 300 t also wins some Jiang decisions with longer handling or slower travel

Scope: 4 conditions (Jiang, Roh–Cha masses × short, mass-dependent handling) × 5 reference fleets (270, 300, 380, 550 t, 270 t + 2 heavy); alignment time was also run at 2.5 and 5 min with six fleets.

Sources: tested levels on “Parameter sources (2)”; Liu's 50/30 m/min speeds Liu et al. (2022) §4.1.1. Results: this study (paper §5.4, §5.5; Supplementary S3).

Fleets sized exactly to the target are lean: under ±50% handling noise most fall below 95%; re-sizing for noise changes the winner, but it still carries alone

On-time share under ±50% handling noise as units are added to K* (one line per fleet; schedules chosen on 20 scenarios and evaluated on 20 others)



- At K*: 89.0–97.4% with short and 79.8–93.1% with mass-dependent handling; only 15 of 66 fleets keep 95%
- Short handling: one extra unit suffices; mass-dependent: Jiang masses need one to three, and none of the 22 Roh–Cha and Liu fleets gets there within three

145 / 360

decided decisions change their winner: under Jiang masses, 270 t with two heavy units replaces one heavy unit (short handling) and 500 t replaces 425 t (mass-dependent)

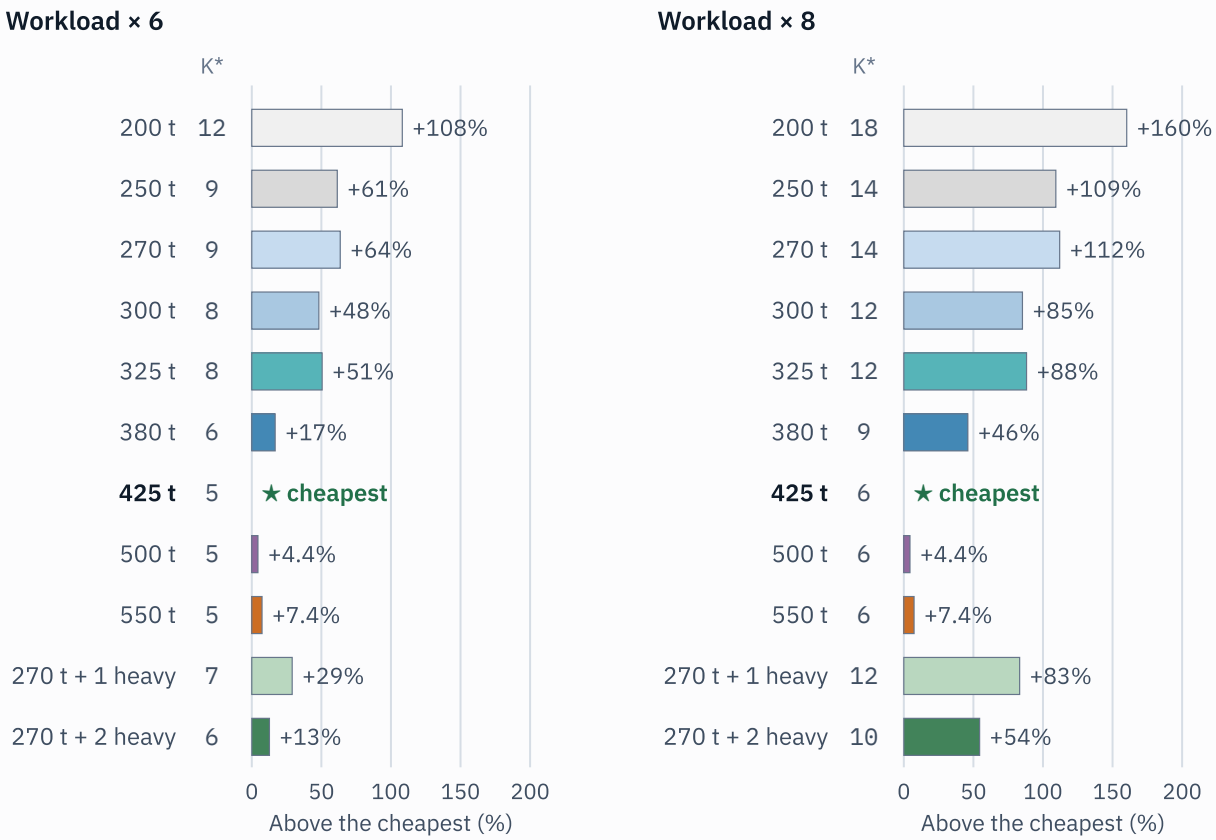
≈ 0 blocks coupled by the winners under noise (≤ 0.1%)

At the noise-robust counts, 42 of 44 fleets keep every block within the 120-min cap in every scenario; the other two exceed it on 0.8% of scenario-days.

Sources: ±50% and 20 + 20 scenarios are our test design; results: this study (paper §5.5; Supplementary Fig. S1, S4).

A second yard from published data gives the same answer: 425 t is cheapest and couples nothing

Price ratio θ 12.4



A Shanghai cruise-ship yard's published week of transfers, distance table and five units of 250–420 t; demand × k = 1–8, about 10·k moves a day, 30 days, 10-h shift. A reconstructed reference case, not a deployment.

- **Sized from scratch** (k = 6, 8): 425 t needs 5 and 6 units (270 t needs 9 and 14) and couples nothing; it is cheapest in 179 of the 180 decisions, by up to 10.6%
- **Adding to the existing five**: they suffice up to twice the demand; from 4× without coupling, light additions never qualify while one or two 420 t units do (at 8× with after-shift work)

The heavy units of the mixes are scheduled as 550 t vehicles and priced at 425 t. In the remaining decision, at six times the demand, 270 t units with two heavy units, coupling 7.6% of blocks, are 0.5% cheaper.

Sources: data Liu et al. (2022), Tables 4–5 and supplementary data S1 (PLOS ONE, CC BY); the 10-h shift and demand multiples are assumptions; decisions cover five price curves × nine values of θ × shift labour and crew-vehicle hours. Results: this study (paper §5.5, Table 7; Supplementary S6).

When does coupling pay? Four situations, and the limits of this study

Four situations where coupling pays or is needed

- **No tier can carry the block**: coupling is necessary (the heavy-tail profile)
- **Near-instant alignment**: at δ = 0 a coupling 300 t fleet wins 43% of one condition's cost scenarios, at 2.5 min none
- **One crew drives a coupled team**: crew-team hours give the lowest low-coupling share (86.9%) and, where a covering tier exists, the only switches inside the calibrated range
- **Capital far dearer than crews**: beyond about 1.4 crew-days per transporter-day, which remotely operated or automated transporters could approach

> **Under the conditions studied, least-cost purchasing favours single-carrier capacity; coupling is kept for exceptional blocks, and redundancy favours a second heavy unit or one more transporter.**

Limitations

- Capital and labour only (no maintenance, energy or resale); capacity loss uses reduced compositions with reassigned starts, so service after an arbitrary failure is unverified
- Four stylised labour measures, not a crew roster; reference mixes' heavy units priced at the smallest tier they need (favouring mixes)
- Stationary demand: seasonal peaks, rental and staged expansion are out of scope
- Static shortest paths, no traffic or conflicts between coupled teams; joint control, load distribution, road width, clearance, gradients and wheel loads unmodelled (width, clearance and wheel loads also limit the heaviest single carriers; net effect untested)
- Process routes (shops, stockyard, dock) and road paths are inputs, not optimised; joint optimisation with the fleet is future work
- Whole-day K* are the smallest counts found, not certified minima
- Excluded: slow-speed and long-handling cases where no candidate meets the specification

Sources: this study (paper §5.6, §6).

Conclusions: answers to the three research questions, and future work

(i) Least-cost fleets rely little on coupling

Under full-shift staffing the least-cost fleet couples at most one block in ten in 91.0% of 1,620 decisions, and in all of them outside the heavy tail. In 13 further conditions, 300 t units with covering heavy units, their number set by the handling-share rule, win 95.3% of 2,340 decisions, all coupling at most one in ten.

(ii) Coupling competes only within a small budget of workload

With sub-additive prices and per-vehicle crews, a large-enough single carrier has a lower fluid cost per block than a coupled team; a light homogeneous fleet is cheaper only while its coupled share stays below the budget, 2–18% of workload under full-shift staffing and the fitted power law. Repeated service and alignment are a median 89.8% of light fleets' extra transporter-time.

(iii) A lateness cap affects both size and composition

Without a cap, fleets qualify by deferring their hardest blocks for hours. With the 120-min cap six sized fleets need one more transporter, and for the Jiang sample and the uniform profile every winner couples at most one block in ten, against 75% of decisions without a cap.

Scope

The conclusions rest on generated daily demand, static shortest paths and coupled teams assumed physically feasible; whole-day counts are the smallest found. Exact single-batch benchmarks certify the least-cost fleet in all 360 shift-labour decisions. The cheapest tonnage is less stable than the kind of fleet, so close purchases should be reported as near-optimal sets, and the coupling budget reassessed when capital-to-crew costs change.

Future work

- Check the physical feasibility of coupled teams under road geometry and load-distribution constraints
- Optimise process routes together with the fleet
- Extend the comparison to staged fleet expansion under seasonal demand

Sources: this study (paper §5.6, §6).

Status: all experiments are complete; the manuscript has been revised after co-author review and is being finalised

Done

Main study: 36 conditions × 11 fleet types

Price and wage calibration

Search depth and boundary re-search

Exact benchmark (single batches)

Day resampling and 30 fresh days

Adversarial bound and evidence grading

Handling noise and robust counts

Lateness cap and on-time targets 90% / 98%

Heavy-block share and extended mixes (16 conditions)

Restricted coupling and alignment time

Manufacturer speeds, one-factor sensitivity, slow speeds scanned to 40

Capacity-removal estimate

Block-by-block decomposition

Second yard

Next

- A final read of the revised manuscript against the result tables
- Archive the released code and result tables with a DOI
- Submit to Ocean Engineering

Paper

Service-constrained fleet sizing for single-carrier and cooperative block transport in shipyards

214,210

scheduling runs, about 2,570 core-hours, all reproducible

Sources: this study's run records and the revised manuscript.

Five rules for a yard planning its transporter fleet

1

Make every frequent block carriable by one transporter

The least-cost fleet is the smallest tier that carries (nearly) every block alone, or light units with a few heavy ones; more capacity buys nothing. This qualifies the advice of Liu et al. (2022).

2

Use the coupling budget with calibrated occupancy

$h^* = \kappa \div (1 + 2\delta / \bar{w})$, about 2–18% under shift labour and $\alpha = 0.84$; with occupancy from covering-fleet schedules, exclude the lighter tier above a fluid cost ratio of 1.10.

3

Serve a thin heavy tail with a few heavy units, not coupling

Heavy units $[K_{cov} \cdot h_H]$, within one unit in 96.6% of the decisions tested.

4

Compare spare capacity with coupling for redundancy

Coupling is indispensable above every tier; in the capacity-removal estimate, least-cost redundancy is a second heavy unit or one more transporter. Failure protection must check each surviving vehicle's capacity and position.

5

Specify a lateness cap; report near-optimal sets

An on-time share alone lets fleets defer their hardest blocks for hours; rankings are often within 1–2%, so report the near-optimal fleets and the decisions a better search could flip.

In one sentence: coupling is a remedy for exceptional blocks, not a purchasing strategy.

Thank you. Questions and comments are welcome.

Sources: rules 1–5 rest on this study's results (paper §5.6); the advice of Liu et al. (2022) is in their discussion.

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